

Digital Electronics & Electrical Fundamentals

ECE249 · Semester 1 & 2 Study Guide

Circuit Laws ▪ Semiconductors ▪ Arduino Interfacing ▪ Logic Gates ▪ Counters

Subject Features:

- Comprehensive coverage of DC circuit laws, network theorems, and semiconductor band theory.
- Solid-state devices: PN diodes, Zener regulation, BJT biasing, and switching operations.
- Embedded systems companion: Arduino UNO architecture, hardware pinouts, and sensor interfacing.
- Combinational logic, Boolean algebra, Karnaugh maps, latches, flip-flops, and synchronous counters.

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Preface & Course Syllabus Correlation

This textbook has been engineered from the ground up to serve as the definitive, university-level curriculum companion and self-study guide for **ECE249: Fundamentals of Electrical Laws, Semiconductor Devices, Digital Electronics, and Arduino Interfacing**.

Course Architectural Philosophy

Digital engineering and embedded systems form the backbone of modern technological infrastructure. Mastery of these fields requires a seamless bridge connecting:

1. **Continuous Physics & DC Circuit Laws:** How electrons move across potential gradients, resistors, diodes, and transistors according to Ohm's Law and Kirchhoff's Laws.
2. **Microcontroller & Sensor Integration:** How continuous physical phenomena (light, sound, heat, distance) are transduced, digitized via ADCs, and processed by embedded platforms like the Arduino UNO.
3. **Discrete Mathematical Foundations:** Positional binary number systems, error-detecting codes, Gray code encoders, and Boolean algebraic minimization using Karnaugh Maps.
4. **Combinational Circuit Synthesis:** Building memoryless processing blocks including adders, subtractors, multiplexers, decoders, encoders, and magnitude comparators.
5. **Sequential State Machines:** Storage latches, edge-triggered flip-flops, master-slave glitch elimination, and shift register counters.

Syllabus Mapping Matrix

Unit	Core Topic Domain	Chapter	Key Highlights
Unit 1	Electrical Laws & Semiconductors	Chapter 1	Ohm's Law, KCL, KVL, VDR, CDR, E_g , Doping, Diode V-I, Rectifiers, BJT CE Mode.
Unit 2	Arduino & Sensor Interfacing	Chapter 2	Analog vs. Digital, ATmega328P, Pinout, ADC, PWM, Active IR, LDR, Ultrasonic, DHT11/22.
Unit 3	Number Systems & Logic Gates	Chapter 3	Base r conversions, BCD, Excess-3, Gray Code, 2's Comp Arithmetic, 7 Gates, De Morgan's, K-Maps.
Unit 4	Combinational Logic Circuits	Chapter 4	Adders, Subtractors, 4:1/8:1 MUX, 1:4 DEMUX, Decoders, Encoders, Comparators.
Unit 5	Sequential Logic Circuits	Chapter 5	Latches, SR/JK/D/T Flip-Flops, Race-Around, Master-Slave, 5-Step FF Conversion.
Unit 6	Sequential Applications	Chapter 6	SISO/SIPO/PISO/PIPO Shift Registers, Ripple UP/DOWN/Mod-N, Synchronous, Ring & Johnson.
Review	Formula Mastery & Practice	Chapter 7 & 8	Master Revision Sheet, 40+ Solved Step-by-Step Numerical & Synthesis Problems.
Exams	University Assessment Papers	Chapter 9	Two Complete 100-Mark Mock Exams (Section A, B, C) with full Solutions & Rubric.

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Fundamentals of Electrical Laws, Semiconductor Devices and its Applications

1.1 Introduction & Chapter Overview

Electrical and electronics engineering rests upon two foundational pillars:

1. **Macroscopic Network Laws:** The fundamental mathematical and physical principles governing how charge, potential, and current distribute through electrical networks (Ohm's Law, Kirchhoff's Laws, and Division Rules).
2. **Microscopic Solid-State Physics:** The quantum and crystalline properties of semiconductor materials that enable controlled rectification, switching, and amplification (Diodes and Bipolar Junction Transistors).

This chapter develops both domains rigorously from first principles, establishing the circuit analysis techniques and semiconductor device physics essential for digital logic and embedded electronics.

1.2 DC Circuit Analysis Fundamentals

1.2.1 Fundamental Quantities: Charge, Current, Voltage, and Power

An electric circuit is an interconnection of electrical elements forming a closed conducting path.

- **Electric Charge (Q):** The intrinsic property of subatomic particles, measured in Coulombs (C). The charge of a single electron is $e = -1.602 \times 10^{-19} \text{ C}$.
- **Electric Current (I):** The time rate of flow of electric charge past a specified cross-section:

$$I(t) = \frac{dQ(t)}{dt} \quad [\text{A} = \text{C} \cdot \text{s}^{-1}] \quad (1.1)$$

Conventional current flows in the direction of positive charge motion (from higher potential to lower potential), opposite to the physical drift of electrons.

- **Voltage / Potential Difference (V):** The energy (W) required to move a unit charge through an element:

$$V = \frac{dW}{dQ} \quad [\text{V} = \text{J} \cdot \text{C}^{-1}] \quad (1.2)$$

- **Electrical Power (P):** The rate at which energy is delivered or absorbed by an element:

$$P = \frac{dW}{dt} = \frac{dW}{dQ} \cdot \frac{dQ}{dt} = V \cdot I \quad [\text{W} = \text{J} \cdot \text{s}^{-1}] \quad (1.3)$$

1.2.2 Ohm's Law

Discovered experimentally by Georg Simon Ohm in 1827, Ohm's Law relates potential difference, current, and resistance in a conducting medium.

Definition: Ohm's Law

At constant physical temperature and environmental conditions, the current (I) flowing through a linear conducting element is directly proportional to the potential difference (V) applied across its terminals.

Ohm's Law Mathematical Formulation

$$V = I \cdot R \iff I = \frac{V}{R} \iff R = \frac{V}{I} \quad (1.4)$$

Where V is potential difference in Volts (V), I is electric current in Amperes (A), and R is resistance in Ohms (Ω).

On a microscopic scale, resistance is determined by material resistivity (ρ), length (L), and cross-sectional area (A):

$$R = \rho \frac{L}{A} = \frac{L}{\sigma A} \quad (1.5)$$

where $\sigma = \frac{1}{\rho}$ is the electrical conductivity of the conductor ($\text{S} \cdot \text{m}^{-1}$).

Common Pitfall: Limits of Ohm's Law

Ohm's Law is **not** a universal law of nature like Newton's laws. It applies strictly to ****ohmic (linear)**** conductors. It ****fails**** for:

- **Non-linear devices:** Diodes, transistors, and vacuum tubes where the V-I curve is exponential or logarithmic.
- **Thermal variations:** Conductors whose resistance changes as Joule heating ($I^2 R$) alters temperature.
- **Non-metallic conductors:** Electrolytes, ionized gases, and super-conductors.

1.2.3 Kirchhoff's Laws

When circuits contain multiple branches, loops, and energy sources, Ohm's law alone is insufficient. Gustav Kirchhoff formulated two fundamental laws rooted in the fundamental conservation laws of physics.

Kirchhoff's Current Law (KCL)

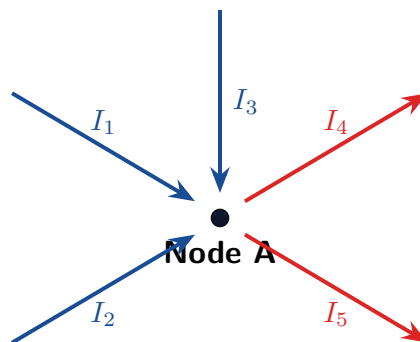
KCL is the direct consequence of the ****Law of Conservation of Electric Charge****. Since electric charge cannot accumulate indefinitely at a junction (node) in steady-state conduction, the rate of charge entering must equal the rate of charge leaving.

Definition: Kirchhoff's Current Law (KCL)

The algebraic sum of currents entering any electrical node (junction) is identically equal to the algebraic sum of currents exiting that node. Alternatively, the algebraic sum of all currents at any node equals zero.

KCL Mathematical Formulation

$$\sum_{k=1}^n I_{k,\text{entering}} = \sum_{j=1}^m I_{j,\text{leaving}} \iff \sum_{k=1}^N I_k = 0 \quad (1.6)$$



$$\text{KCL at Node A: } I_1 + I_2 + I_3 = I_4 + I_5 \implies I_1 + I_2 + I_3 - I_4 - I_5 = 0$$

Kirchhoff's Voltage Law (KVL)

KVL is the direct consequence of the ****Law of Conservation of Energy****. Moving a unit test charge around any closed path and returning to the starting point results in zero net change in electrical potential energy.

Definition: Kirchhoff's Voltage Law (KVL)

The algebraic sum of all branch voltages (potential rises and potential drops) around any closed loop or mesh in an electric circuit is identically zero.

KVL Mathematical Formulation

$$\sum_{k=1}^N V_{k,\text{rise}} = \sum_{j=1}^M V_{j,\text{drop}} \iff \sum_{k=1}^{\text{Total}} V_k = 0 \quad (1.7)$$

Exam Tip: Standard Sign Convention for Loop Analysis

When traversing a closed circuit loop in a chosen direction (clockwise or counter-clockwise):

- Traversing from a **negative (-) to positive (+)** terminal across a source or element represents a **Voltage Rise (+V)**.
- Traversing from a **positive (+) to negative (-)** terminal represents a **Voltage**

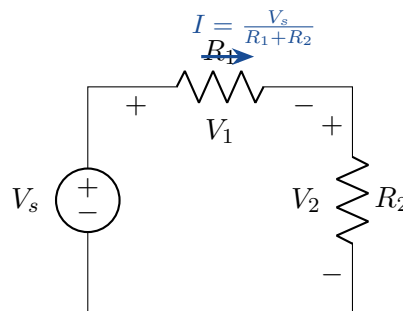
Drop ($-V$).

- For a resistor carrying current I in the direction of traversal, the voltage drop is $-I \cdot R$. If traversing against the current, it is $+I \cdot R$.

1.2.4 Circuit Division Rules

Voltage Division Rule (VDR)

The Voltage Division Rule applies exclusively to **series-connected circuits**. In a series circuit, the identical current I passes through all elements, but the total supply voltage V_s divides across individual resistors proportional to their resistance values.



Voltage Division Rule Formula

For two resistors R_1 and R_2 in series with a source V_s :

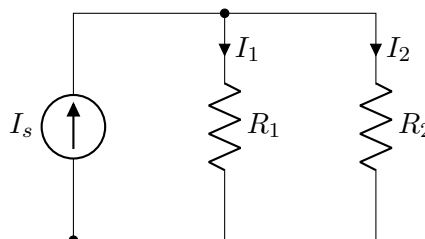
$$V_1 = V_s \left(\frac{R_1}{R_1 + R_2} \right), \quad V_2 = V_s \left(\frac{R_2}{R_1 + R_2} \right) \quad (1.8)$$

Generalizing for N resistors in series:

$$V_k = V_s \left(\frac{R_k}{\sum_{i=1}^N R_i} \right) \quad (1.9)$$

Current Division Rule (CDR)

The Current Division Rule applies exclusively to **parallel-connected circuits**. In a parallel circuit, the voltage across each parallel branch is identical (V_p), but the total entering current I_s divides among the parallel branches inversely proportional to their branch resistances.



Current Division Rule Formula

For two resistors R_1 and R_2 connected in parallel with total current I_s :

$$I_1 = I_s \left(\frac{R_2}{R_1 + R_2} \right), \quad I_2 = I_s \left(\frac{R_1}{R_1 + R_2} \right) \quad (1.10)$$

Common Pitfall: Current Division with Three or More Branches

The formula $I_1 = I_s \frac{R_2}{R_1 + R_2}$ is valid **only for exactly two parallel branches**. For three or more branches (R_1, R_2, R_3), the branch current must be evaluated using branch conductance:

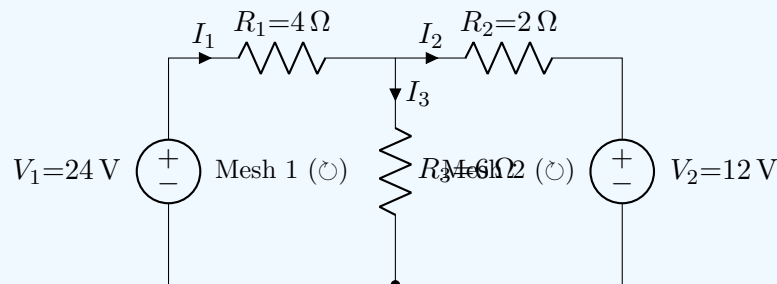
$$I_k = I_s \left(\frac{G_k}{\sum G_i} \right) = I_s \left(\frac{\frac{1}{R_k}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \right) \quad (1.11)$$

Attempting to use $I_1 = I_s \frac{R_2 + R_3}{R_1 + R_2 + R_3}$ is a severe mathematical error.

1.2.5 Worked Examples: DC Circuit Analysis

Worked Example: Multi-Loop DC Circuit Analysis using KVL & KCL

Consider the two-mesh DC network shown below. Find the branch currents I_1, I_2, I_3 and the voltage across the 6Ω resistor.



Step 1: Given Data

- Sources: $V_1 = 24\text{ V}$, $V_2 = 12\text{ V}$
- Resistors: $R_1 = 4\Omega$, $R_2 = 2\Omega$, $R_3 = 6\Omega$

Step 2: Formulate KCL at Central Node At the top central node:

$$I_1 = I_2 + I_3 \implies I_3 = I_1 - I_2 \quad (1.12)$$

Step 3: Apply KVL around Mesh 1 (Clockwise)

$$\begin{aligned} +V_1 - I_1 R_1 - I_3 R_3 &= 0 \\ 24 - 4I_1 - 6(I_1 - I_2) &= 0 \\ 24 - 10I_1 + 6I_2 &= 0 \implies 10I_1 - 6I_2 = 24 \quad \text{--- (Equation 1)} \end{aligned}$$

Step 4: Apply KVL around Mesh 2 (Clockwise)

$$\begin{aligned} +I_3 R_3 - I_2 R_2 - V_2 &= 0 \\ 6(I_1 - I_2) - 2I_2 - 12 &= 0 \\ 6I_1 - 8I_2 &= 12 \implies 3I_1 - 4I_2 = 6 \quad \text{--- (Equation 2)} \end{aligned}$$

Step 5: Solve System of Linear Equations From Equation 1: $5I_1 - 3I_2 = 12 \Rightarrow I_1 = \frac{12+3I_2}{5}$. Substitute into Equation 2:

$$3\left(\frac{12+3I_2}{5}\right) - 4I_2 = 6$$

$$\frac{36 + 9I_2 - 20I_2}{5} = 6$$

$$36 - 11I_2 = 30 \Rightarrow 11I_2 = 6 \Rightarrow I_2 = \frac{6}{11} \approx 0.545 \text{ A}$$

Back-substituting for I_1 :

$$I_1 = \frac{12 + 3(0.5455)}{5} = \frac{13.636}{5} \approx 2.727 \text{ A}$$

Current through R_3 :

$$I_3 = I_1 - I_2 = 2.7273 - 0.5455 = 2.182 \text{ A}$$

Voltage across R_3 :

$$V_{R_3} = I_3 \cdot R_3 = 2.1818 \times 6 = 13.09 \text{ V}$$

Final Answers:

$$I_1 = 2.73 \text{ A}, \quad I_2 = 0.55 \text{ A}, \quad I_3 = 2.18 \text{ A}, \quad V_{R_3} = 13.09 \text{ V}$$

1.3 Basics of Semiconductor Physics

1.3.1 Energy Band Theory of Solids

In an isolated atom, electron energy levels are discrete and sharply quantized. When atoms condense to form a periodic crystalline solid (such as Silicon or Germanium), their atomic wavefunctions overlap. By the **Pauli Exclusion Principle**, each degenerate energy level splits into a band of closely spaced quantum energy states.

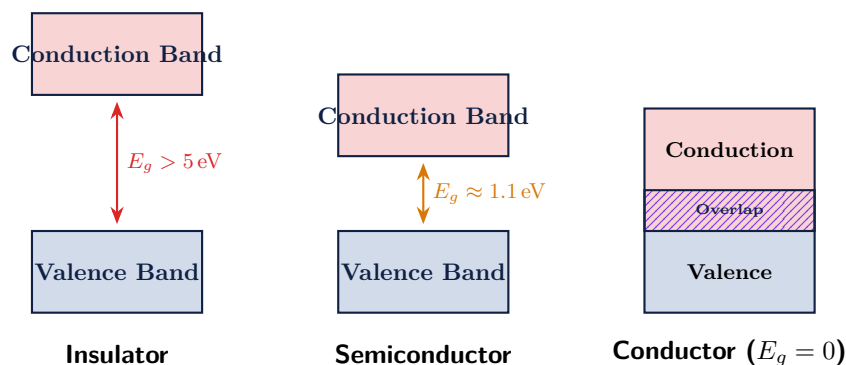


Figure 1.1: Energy band classification of materials based on the forbidden band gap E_g .

- **Valence Band (VB):** The highest range of electron energies in which electrons are normally present at absolute zero ($T = 0 \text{ K}$).

- **Conduction Band (CB):** The lowest unoccupied band above the valence band. Electrons excited into this band are free to drift under an electric field.
- **Forbidden Energy Gap (E_g):** The energy gap between the top of the valence band (E_v) and the bottom of the conduction band (E_c). No stable quantum states exist in this region.
 - **Insulators:** Large gap ($E_g > 5 \text{ eV}$); thermal energy cannot excite electrons across the gap.
 - **Semiconductors:** Moderate gap ($E_g(\text{Si}) = 1.12 \text{ eV}$, $E_g(\text{Ge}) = 0.72 \text{ eV}$ at $T = 300 \text{ K}$).
 - **Conductors:** Overlapping bands ($E_g = 0$); abundant free electrons available even at 0 K .

1.3.2 Intrinsic Semiconductors

An intrinsic semiconductor is chemically pure with zero significant crystal lattice defects or foreign impurity atoms.

- **Crystal Structure:** Group IV elements (Silicon, Germanium) have 4 valence electrons. In a diamond cubic crystal, each atom forms four shared covalent bonds with four neighboring atoms.
- **Behavior at 0 K :** All covalent bonds are intact. The valence band is completely full, the conduction band is completely empty, and the crystal acts as a perfect **electrical insulator**.
- **Thermal Generation at $T > 0 \text{ K}$:** Ambient thermal energy breaks covalent bonds, exciting electrons into the conduction band and leaving behind vacant covalent positions called **Holes**.
- **Concept of the Hole:** A hole behaves as an effective mobile charge carrier with positive elementary charge ($+q = 1.602 \times 10^{-19} \text{ C}$) and its own effective mass and mobility (μ_p).
- **Carrier Equality:** In an intrinsic semiconductor, electrons and holes are generated in pairs:

$$n = p = n_i \quad (1.13)$$

where n_i is the intrinsic carrier concentration ($n_i \approx 1.5 \times 10^{10} \text{ cm}^{-3}$ for Silicon at 300 K).

1.3.3 Extrinsic Semiconductors & Doping

Because intrinsic semiconductors have low conductivity at room temperature, controlled amounts of specific impurity atoms are added via **Doping** (typically 1 impurity atom per 10^6 to 10^8 host atoms).

N-Type Semiconductors (Donor Doping)

- **Dopants:** Pentavalent atoms from Group V (Phosphorus P, Arsenic As, Antimony Sb) possessing 5 valence electrons.
- **Lattice Incorporation:** Four valence electrons form covalent bonds with adjacent Silicon atoms; the fifth electron is loosely bound ($E_d \approx 0.05 \text{ eV}$ below E_c) and easily ionizes at room temperature to enter the conduction band.
- **Charge Carriers:**
 - **Majority Carriers:** Free electrons ($n_n \approx N_D$, where N_D is donor concentration).
 - **Minority Carriers:** Thermally generated holes ($p_n = \frac{n_i^2}{N_D}$).
- **Space Charge:** Doped atoms become fixed, immobile positive ions (P^+). The overall material remains **electrically neutral**.

P-Type Semiconductors (Acceptor Doping)

- **Dopants:** Trivalent atoms from Group III (Boron B, Gallium Ga, Indium In) possessing 3 valence electrons.
- **Lattice Incorporation:** Three valence electrons form bonds with neighboring Silicon atoms, leaving a missing electron (vacancy / hole). This vacancy accepts an electron from an adjacent bond ($E_a \approx 0.05$ eV above E_v), creating a free hole in the valence band.
- **Charge Carriers:**
 - **Majority Carriers:** Holes ($p_p \approx N_A$, where N_A is acceptor concentration).
 - **Minority Carriers:** Thermally generated electrons ($n_p = \frac{n_i^2}{N_A}$).
- **Space Charge:** Doped atoms become fixed, immobile negative ions (B^-). The overall material remains **electrically neutral**.

Mass-Action Law

Under thermal equilibrium conditions, the product of electron and hole concentrations in any semiconductor (intrinsic or extrinsic) is constant:

$$n \cdot p = n_i^2 \quad (1.14)$$

1.4 PN Junction Diode

1.4.1 PN Junction Formation and Depletion Region

When P-type and N-type semiconductor regions are joined in a continuous crystal lattice:

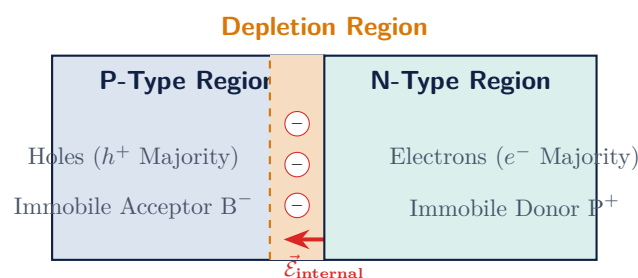


Figure 1.2: Cross-section of an open-circuit PN junction in thermal equilibrium showing the space-charge depletion layer.

1. **Diffusion Process:** Due to the large carrier concentration gradient, majority holes from the P-side diffuse across the junction into the N-side, while majority electrons from the N-side diffuse into the P-side.
2. **Recombination & Space Charge Formation:** When diffusing electrons and holes cross the junction, they recombine near the interface. This uncovers the fixed, ionized dopant atoms: immobile negative acceptor ions (B^-) on the P-side, and immobile positive donor ions (P^+) on the N-side.
3. **Built-in Potential Barrier (V_b):** The layer of uncovered ions forms the **Depletion Region** (depleted of mobile carriers). The resulting internal electric field $\vec{\mathcal{E}}$ points from N

to P, opposing further diffusion until equilibrium is reached:

$$V_b = V_T \ln \left(\frac{N_A N_D}{n_i^2} \right) \quad (1.15)$$

where $V_T = \frac{k_B T}{q} \approx 25.9 \text{ mV}$ at 300 K.

- Silicon Barrier Potential: $V_b \approx 0.7 \text{ V}$
- Germanium Barrier Potential: $V_b \approx 0.3 \text{ V}$

1.4.2 Biasing Modes of the PN Junction

Biasing Mode	Forward Bias ($V_D > 0$)	Reverse Bias ($V_D < 0$)
External Connection	P-terminal connected to (+), N-terminal to (-).	P-terminal connected to (-), N-terminal to (+).
Internal Field Effect	Opposes built-in barrier field $\mathcal{E}$.	Reinforces built-in barrier field $\mathcal{E}$.
Depletion Layer Width	Significantly decreases / narrows .	Significantly increases / widens .
Potential Barrier	Reduces to $(V_b - V_D)$.	Increases to $(V_b + V_D)$.
Conduction Mechanism	Exponential diffusion of majority carriers.	Negligible drift of minority carriers (I_0).
Equivalent Circuit	Low forward resistance ($r_f \approx 10 \Omega$) in series with V_k .	High resistance ($R_r > 1 \text{ M}\Omega$) or open circuit.

Table 1.1: Comparison of Forward and Reverse Biasing in a PN Junction Diode.

1.4.3 V-I Characteristics & Shockley Diode Equation

The relationship between diode terminal current (I) and applied voltage (V) is described by the ****Shockley Equation****:

Shockley Ideal Diode Equation

$$I = I_0 \left(e^{\frac{V}{\eta V_T}} - 1 \right) \quad (1.16)$$

Where:

- I_0 = Reverse saturation current (10^{-12} A for Si, 10^{-6} A for Ge)
- V = Applied terminal voltage across diode
- η = Ideality factor ($\eta = 1$ for Ge, $\eta \approx 1-2$ for Si)
- $V_T = \frac{k_B T}{q}$ = Thermal voltage ($\approx 26 \text{ mV}$ at 300 K)

****Key Characteristic Parameters:****

- **Knee Voltage / Cut-in Voltage (V_k):** Forward voltage at which current begins to increase exponentially (0.7 V for Si, 0.3 V for Ge).
- **Static (DC) Resistance (R_D):** The ratio of DC voltage to DC current at a fixed Q-point: $R_D = \frac{V_Q}{I_Q}$.
- **Dynamic (AC) Resistance (r_d):** The reciprocal of the slope of the V-I curve at the operating point:

$$r_d = \frac{dV}{dI} = \frac{\eta V_T}{I_D} \quad (1.17)$$

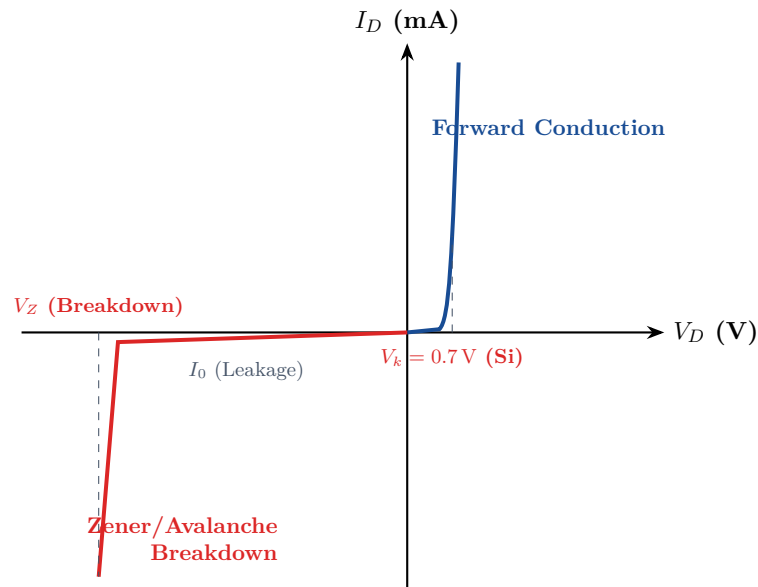


Figure 1.3: Complete V-I characteristics curve of a Silicon PN junction diode.

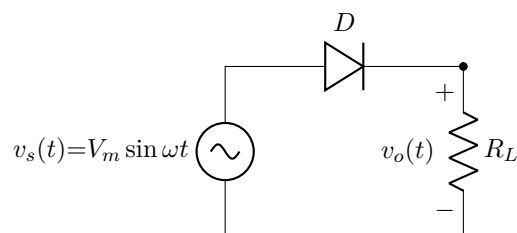
- **Breakdown Voltage (V_{BR}):** Critical reverse voltage where large reverse currents flow via ****Zener Breakdown**** (quantum mechanical tunneling in heavily doped diodes, $< 6\text{ V}$) or ****Avalanche Breakdown**** (carrier impact ionization in lightly doped diodes, $> 6\text{ V}$).

1.5 Diode Applications: Rectifier Circuits

Rectification is the conversion of alternating current (AC) into unidirectional direct current (DC).

1.5.1 Half-Wave Rectifier (HWR)

A Half-Wave Rectifier utilizes a single diode to conduct only during the positive half-cycle of the input AC voltage.



- **Positive Half-Cycle ($0 \leq \omega t \leq \pi$):** Anode is positive; diode is forward-biased (ON). Current flows through R_L , yielding $v_o(t) = V_m \sin \omega t$.
- **Negative Half-Cycle ($\pi \leq \omega t \leq 2\pi$):** Anode is negative; diode is reverse-biased (OFF). Current is zero, yielding $v_o(t) = 0$.

Half-Wave Rectifier Performance Parameters

$$V_{dc} = \frac{1}{2\pi} \int_0^\pi V_m \sin \omega t d(\omega t) = \frac{V_m}{\pi} \approx 0.318V_m \quad (1.18)$$

$$I_{dc} = \frac{I_m}{\pi} \quad (1.19)$$

$$V_{rms} = \sqrt{\frac{1}{2\pi} \int_0^\pi (V_m \sin \omega t)^2 d(\omega t)} = \frac{V_m}{2} = 0.5V_m \quad (1.20)$$

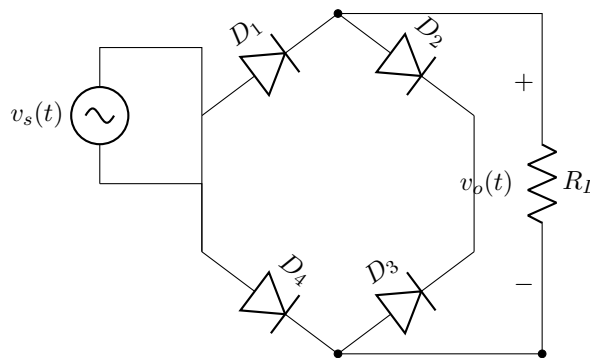
$$\text{Ripple Factor } \gamma = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2 - 1} = \sqrt{\left(\frac{V_m/2}{V_m/\pi}\right)^2 - 1} = \sqrt{\frac{\pi^2}{4} - 1} \approx 1.21 \quad (1.21)$$

$$\text{Max Efficiency } \eta_{max} = \frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 R_L}{I_{rms}^2 (r_f + R_L)} \approx \frac{4}{\pi^2} \approx 40.6\% \quad (1.22)$$

$$\text{Peak Inverse Voltage (PIV)} = V_m \quad (1.23)$$

1.5.2 Full-Wave Bridge Rectifier

The Bridge Rectifier uses four diodes arranged in a bridge loop to conduct current through the load in the same direction during both AC half-cycles, without requiring a center-tapped transformer.



- **Positive Half-Cycle:** Terminal top is (+), bottom is (-). Diodes D_1 and D_3 are Forward-Biased (ON); D_2 and D_4 are Reverse-Biased (OFF). Current enters top of R_L .
- **Negative Half-Cycle:** Terminal top is (-), bottom is (+). Diodes D_2 and D_4 are Forward-Biased (ON); D_1 and D_3 are Reverse-Biased (OFF). Current enters top of R_L in the identical direction.

Full-Wave Bridge Rectifier Performance Parameters

$$V_{dc} = \frac{2V_m}{\pi} \approx 0.636V_m \quad (1.24)$$

$$I_{dc} = \frac{2I_m}{\pi} \quad (1.25)$$

$$V_{rms} = \frac{V_m}{\sqrt{2}} \approx 0.707V_m \quad (1.26)$$

$$\text{Ripple Factor } \gamma = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2 - 1} = \sqrt{\left(\frac{V_m/\sqrt{2}}{2V_m/\pi}\right)^2 - 1} = \sqrt{\frac{\pi^2}{8} - 1} \approx 0.482 \quad (1.27)$$

$$\text{Max Efficiency } \eta_{max} = \frac{8}{\pi^2} \approx 81.2\% \quad (1.28)$$

$$\text{Peak Inverse Voltage (PIV)} = V_m \quad (\text{Bridge}) \quad [2V_m \text{ for Center-Tapped}] \quad (1.29)$$

Parameter	Half-Wave	Center-Tapped Full-Wave	Bridge Full-Wave
Number of Diodes	1	2	4
Transformer Center-Tap Required	No	Yes	No
DC Output Voltage (V_{dc})	$\frac{V_m}{\pi} \approx 0.318V_m$	$\frac{2V_m}{\pi} \approx 0.636V_m$	$\frac{2V_m}{\pi} \approx 0.636V_m$
RMS Output Voltage (V_{rms})	$\frac{V_m}{\sqrt{2}} = 0.707V_m$	$\frac{V_m}{\sqrt{2}} \approx 0.707V_m$	$\frac{V_m}{\sqrt{2}} \approx 0.707V_m$
Ripple Factor (γ)	1.21	0.482	0.482
Ripple Frequency (f_{ripple})	f_{in}	$2f_{in}$	$2f_{in}$
Max Efficiency (η_{max})	40.6%	81.2%	81.2%
Peak Inverse Voltage (PIV)	V_m	$2V_m$	V_m
Transformer Utilization Factor	0.287	0.693	0.812

Table 1.2: Comprehensive performance comparison of rectifier topologies.

Worked Example: Full-Wave Bridge Rectifier Design Calculation

A Full-Wave Bridge Rectifier is supplied by a sinusoidal transformer secondary voltage $v_s(t) = 230\sqrt{2}\sin(100\pi t)$ V with a load resistance $R_L = 500\Omega$. Assuming practical silicon diodes each having a forward drop $V_D = 0.7$ V:

1. Find the peak voltage across the load $V_{L,peak}$.
2. Calculate the DC output voltage V_{dc} and DC load current I_{dc} .
3. Determine the ripple voltage V_{ac} and the ripple factor.
4. What is the minimum PIV rating required for each diode?

Step 1: Given Data

- Peak secondary voltage: $V_m = 230\sqrt{2} \approx 325.27$ V
- Frequency: $\omega = 100\pi \Rightarrow f = \frac{100\pi}{2\pi} = 50$ Hz
- Load: $R_L = 500\Omega$, Diode drop: $V_D = 0.7$ V (2 conducting diodes in series during each half cycle)

Step 2: Peak Load Voltage

$$V_{L,peak} = V_m - 2V_D = 325.27 - 2(0.7) = 323.87 \text{ V}$$

Step 3: DC Voltage and DC Current

$$V_{dc} = \frac{2V_{L,peak}}{\pi} = \frac{2 \times 323.87}{\pi} \approx 206.18 \text{ V}$$

$$I_{dc} = \frac{V_{dc}}{R_L} = \frac{206.18}{500 \Omega} = 0.4124 \text{ A} = 412.4 \text{ mA}$$

Step 4: RMS Voltage, Ripple Voltage and Ripple Factor

$$V_{rms} = \frac{V_{L,peak}}{\sqrt{2}} = \frac{323.87}{\sqrt{2}} \approx 229.01 \text{ V}$$

$$V_{ac} = \sqrt{V_{rms}^2 - V_{dc}^2} = \sqrt{(229.01)^2 - (206.18)^2} = \sqrt{52445.58 - 42510.19} = 99.68 \text{ V}$$

$$\gamma = \frac{V_{ac}}{V_{dc}} = \frac{99.68}{206.18} \approx 0.483$$

Step 5: Peak Inverse Voltage (PIV)

$$PIV = V_m - V_D = 325.27 - 0.7 = 324.57 \text{ V}$$

Final Answers:

$$V_{L,peak} = 323.87 \text{ V}, \quad V_{dc} = 206.18 \text{ V}, \quad I_{dc} = 412.4 \text{ mA}, \quad \gamma = 0.483, \quad PIV \geq 324.6 \text{ V}$$

1.6 Bipolar Junction Transistor (BJT)

1.6.1 Construction and Physical Mechanism

The Bipolar Junction Transistor (BJT) is a three-terminal semiconductor device containing two back-to-back PN junctions (J_E and J_C). It is termed *bipolar* because conduction involves both majority and minority carriers (electrons and holes).

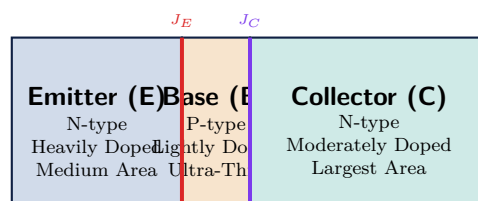


Figure 1.4: Physical structural proportions and doping hierarchy of an NPN Bipolar Junction Transistor.

- **Emitter (E):** Heavily doped (N_D^{++}) to emit a massive quantity of majority carriers into the base. Moderate physical size.
- **Base (B):** Ultra-thin ($\sim 1 \mu\text{m}$) and lightly doped (N_A^-) to minimize electron-hole recombination as carriers traverse toward the collector.
- **Collector (C):** Moderately doped (N_D^+) and physically the largest region to facilitate thermal heat dissipation generated by collected carriers.

1.6.2 Fundamental Current Relationships

Applying Kirchhoff's Current Law to the BJT package:

BJT Fundamental Current Equation

$$I_E = I_B + I_C \quad (1.30)$$

In active operation, because the base is lightly doped and extremely thin, approximately 95–99% of emitted carriers traverse directly to the collector. Thus:

$$I_B \approx (1\text{--}5\%) \text{ of } I_E, \quad I_C \approx (95\text{--}99\%) \text{ of } I_E \implies I_E \approx I_C \quad (1.31)$$

1.6.3 BJT Operating Modes

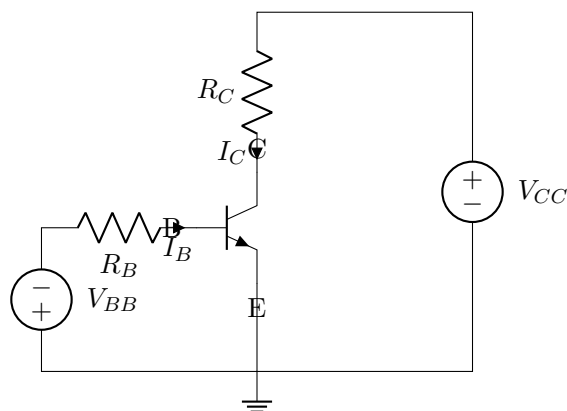
The biasing polarities of the Emitter-Base Junction (J_E) and Collector-Base Junction (J_C) establish four distinct operational regions:

Mode of Operation	J_E Bias (EBJ)	J_C Bias (CBJ)	Primary Application
Cut-off Mode	Reverse	Reverse	Open Electronic Switch (OFF)
Active Mode	Forward	Reverse	Linear Analog Signal Amplifier
Saturation Mode	Forward	Forward	Closed Electronic Switch (ON)
Inverted Mode	Reverse	Forward	Low-gain specialized applications

Table 1.3: BJT Operational Modes and Biasing Requirements.

1.6.4 Common Emitter (CE) Configuration & Characteristics

The Common Emitter (CE) configuration is the most widely utilized BJT circuit arrangement due to its simultaneous high current gain and high voltage gain.



1. CE Input Characteristics (I_B vs. V_{BE})

Plots base current I_B (μA) versus base-emitter voltage V_{BE} (V) at fixed values of collector-emitter voltage V_{CE} .

- Resembles a forward-biased diode characteristic with a knee voltage at $V_{BE} \approx 0.7\text{ V}$ for Silicon.
- Increasing V_{CE} slightly shifts the curve to the right due to base-width modulation (Early effect).

2. CE Output Characteristics (I_C vs. V_{CE})

Plots collector current I_C (mA) versus collector-emitter voltage V_{CE} (V) for constant base current steps I_B .

- **Cut-off Region** ($I_B \leq 0$): Both junctions reverse-biased. $I_C = I_{CEO} \approx 0$. Transistor behaves as an open switch.
- **Saturation Region** ($V_{CE} < V_{CE(sat)} \approx 0.2\text{ V}$): Both junctions forward-biased. I_C is limited by external collector resistance R_C , not by I_B . Transistor behaves as a closed switch.
- **Active Region** ($V_{CE} > 0.2\text{ V}$): J_E forward-biased, J_C reverse-biased. I_C is nearly horizontal and controlled linearly by I_B :

$$I_C = \beta I_B \quad (1.32)$$

1.6.5 Current Amplification Factors: α and β

- **Common Base Current Gain (α):**

$$\alpha = \frac{I_C}{I_E} \quad (\text{Typical range: } 0.95 \text{ to } 0.998) \quad (1.33)$$

- **Common Emitter Current Gain (β or h_{FE}):**

$$\beta = \frac{I_C}{I_B} \quad (\text{Typical range: } 20 \text{ to } 500) \quad (1.34)$$

Mathematical Relationship Between α and β

From $I_E = I_B + I_C$, dividing through by I_C :

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + 1 \implies \frac{1}{\alpha} = \frac{1}{\beta} + 1 = \frac{\beta + 1}{\beta} \quad (1.35)$$

Solving for β and α :

$$\beta = \frac{\alpha}{1 - \alpha} \iff \alpha = \frac{\beta}{1 + \beta} \quad (1.36)$$

Worked Example: BJT CE Bias Point & Gain Calculation

An NPN transistor in CE configuration operates in the active region with a base current $I_B = 40\text{ }\mu\text{A}$ and current gain $\beta = 120$.

1. Calculate the collector current I_C and emitter current I_E .
2. Determine the common-base current gain α .
3. If $V_{CC} = 12\text{ V}$ and $R_C = 1.8\text{ k}\Omega$, compute the collector-emitter voltage V_{CE} and confirm whether the transistor operates in the active mode.

Step 1: Given Data

- $I_B = 40\text{ }\mu\text{A} = 0.040\text{ mA}$, $\beta = 120$, $V_{CC} = 12\text{ V}$, $R_C = 1.8\text{ k}\Omega$

Step 2: Calculate I_C and I_E

$$I_C = \beta I_B = 120 \times 40 \mu\text{A} = 4800 \mu\text{A} = 4.8 \text{ mA}$$

$$I_E = I_B + I_C = 0.040 \text{ mA} + 4.8 \text{ mA} = 4.84 \text{ mA}$$

Step 3: Calculate α

$$\alpha = \frac{\beta}{1 + \beta} = \frac{120}{121} \approx 0.9917$$

Step 4: Calculate V_{CE} and Verify Active Mode Applying KVL to the collector-emitter loop:

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C = 12 \text{ V} - (4.8 \text{ mA} \times 1.8 \text{ k}\Omega) = 12 - 8.64 = 3.36 \text{ V}$$

Since $V_{CE} = 3.36 \text{ V} > V_{CE(sat)} \approx 0.2 \text{ V}$, the collector-base junction remains reverse-biased ($V_{CB} = V_{CE} - V_{BE} = 3.36 - 0.7 = 2.66 \text{ V} > 0$), confirming stable ****Active Region Operation****.

Final Answers:

$$I_C = 4.8 \text{ mA}, \quad I_E = 4.84 \text{ mA}, \quad \alpha = 0.9917, \quad V_{CE} = 3.36 \text{ V (Active Mode)}$$

1.7 Summary & Chapter Review

Quick Revision: Unit 1 Essential Review Takeaways

- **Ohm's Law:** $V = IR$; valid for ohmic materials at constant temperature.
- **Kirchhoff's Current Law (KCL):** $\sum I = 0$ (Conservation of Charge at a node).
- **Kirchhoff's Voltage Law (KVL):** $\sum V = 0$ (Conservation of Energy in a closed loop).
- **VDR & CDR:** $V_1 = V_s \frac{R_1}{R_1 + R_2}$ (series); $I_1 = I_s \frac{R_2}{R_1 + R_2}$ (parallel).
- **Semiconductor Gaps:** Insulator ($> 5 \text{ eV}$), Silicon (1.12 eV), Germanium (0.72 eV), Conductor (0 eV).
- **Extrinsic Doping:** N-type (Pentavalent: majority electrons, $n \approx N_D$), P-type (Trivalent: majority holes, $p \approx N_A$). Mass action: $n \cdot p = n_i^2$.
- **PN Diode Knee Voltages:** Silicon = 0.7 V , Germanium = 0.3 V .
- **Rectification Efficiency:** Half-Wave $\eta_{max} = 40.6\%$, Full-Wave $\eta_{max} = 81.2\%$. Ripple factor: HWR = 1.21 , FWR = 0.482 .
- **BJT Relations:** $I_E = I_B + I_C$, $I_C = \beta I_B$, $\beta = \frac{\alpha}{1 - \alpha}$, Active mode requires J_E forward and J_C reverse biased.

Introduction of Arduino, Signals, and Sensor Interfacing

2.1 Introduction & Engineering Relevance

The physical world is inherently analog—temperatures fluctuate continuously, acoustic pressures oscillate smoothly, and illumination levels vary over an infinite spectrum. Conversely, digital microprocessors and microcontrollers operate exclusively on discrete binary states (**0** and **1**). This chapter examines the bridge between these domains: signal classification, the architecture and peripheral subsystems of the **Arduino UNO (ATmega328P)** platform, and the operational physics of primary electronic sensors (IR, LDR, Ultrasonic, and DHT11/22).

2.2 Signal Characterization: Analog vs. Digital

2.2.1 Analog Signals

An analog signal is a time-varying, continuous physical quantity whose amplitude can assume an infinite number of values within a specified range.

- **Continuous in Time and Amplitude:** At every infinitesimal instant t , there exists a corresponding defined potential $v(t)$.
- **Mathematical Representation:** Often represented as Fourier series combinations of sinusoidal waveforms:

$$v(t) = V_m \sin(\omega t + \phi) \quad (2.1)$$

- **Vulnerability to Noise:** Any external electromagnetic interference (EMI) directly corrupts the signal amplitude, making perfect noise separation impossible without filtering.

2.2.2 Digital Signals

A digital signal is a discrete representation of information, constrained to a finite set of predetermined voltage levels (typically binary: HIGH and LOW).

- **Discrete States:** In TTL (Transistor-Transistor Logic) systems:
 - **Logic HIGH (1):** Defined between +3.5 V and +5.0 V.
 - **Logic LOW (0):** Defined between 0 V and +0.8 V.
 - **Indeterminate Region:** 0.8 V to 2.0 V (forbidden transition zone).
- **Noise Immunity:** Because receivers only distinguish whether a voltage is above or below a threshold, minor noise fluctuations do not alter the binary data value.

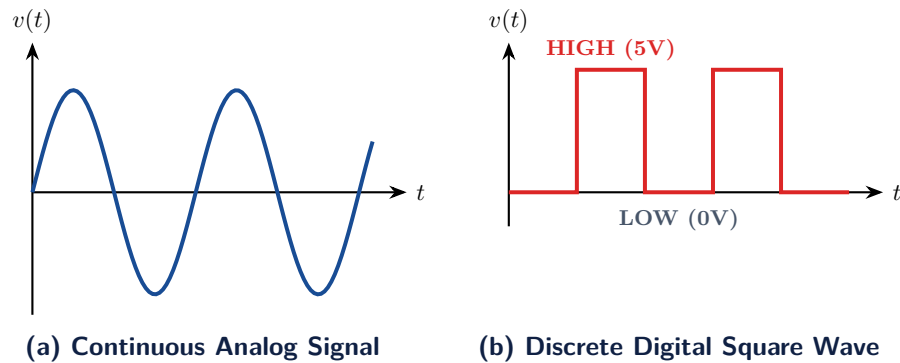


Figure 2.1: Comparison between (a) continuous analog and (b) discrete digital signals.

Parameter	Analog Signal	Digital Signal
Nature	Continuous in time and amplitude.	Discrete in time and quantized in amplitude.
Information Representation	Continuous physical voltage or current.	Binary bit strings (0s and 1s).
Possible Values	Infinite values within dynamic range.	Exactly two states (HIGH / LOW).
Noise Immunity	Low (noise directly alters signal data).	High (noise rejected by threshold margins).
Storage & Transmission	Difficult without signal degradation.	Easily stored (Flash, RAM, EEPROM) without loss.
Processing Hardware	Operational amplifiers, passive RLC filters.	Microcontrollers, DSPs, FPGA, Logic gates.
Real-World Examples	Natural human voice, LDR light level, thermocouple.	Computer bus data, square clock, digital switch.

Table 2.1: Systematic comparison between Analog and Digital Signals.

2.2.3 Comparative Analysis: Analog vs. Digital Signals

2.3 The Arduino UNO Architecture & Hardware Ecosystem

2.3.1 Microcontroller Specifications (ATmega328P)

The **Arduino UNO** is an open-source microcontroller development board centered on the 8-bit **Microchip ATmega328P** AVR RISC processor.

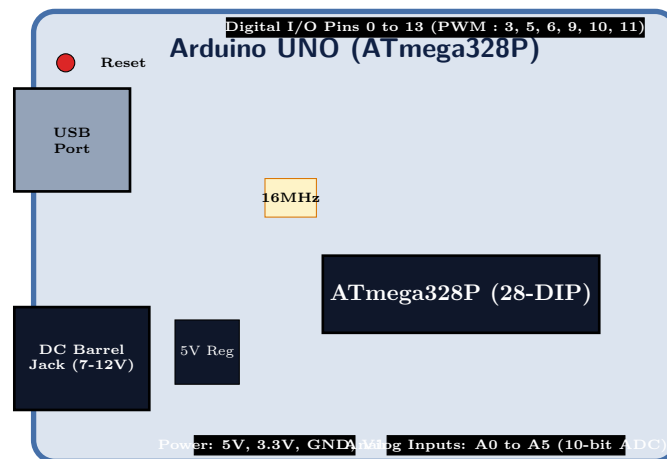


Figure 2.2: Architectural layout and peripheral subsystems of the Arduino UNO board.

Arduino UNO Technical Specification Matrix

- **Microcontroller:** Microchip ATmega328P (8-bit AVR Enhanced RISC Architecture)
- **Operating Voltage:** 5.0 V DC
- **Recommended Input Voltage (VIN / Barrel Jack):** 7 V to 12 V DC (Absolute limits: 6–20 V)
- **Digital I/O Pins:** 14 total (Pins 0 to 13)
- **PWM (Pulse Width Modulation) Channels:** 6 pins (Pins 3, 5, 6, 9, 10, 11 marked with ~)
- **Analog Input Channels:** 6 channels (Pins A0 to A5, 10-bit SAR ADC)
- **DC Current per I/O Pin:** 20 mA recommended (40 mA absolute maximum)
- **DC Current for 3.3V Pin:** 50 mA maximum
- **Flash Memory:** 32 kB (of which 0.5 kB is reserved by bootloader)
- **SRAM (Static RAM):** 2 kB (volatile data storage)
- **EEPROM (Electrically Erasable PROM):** 1 kB (non-volatile storage)
- **Clock Frequency:** 16 MHz (governed by quartz crystal oscillator)

2.3.2 Pin Configuration and Functional Descriptions

1. Power Distribution Subsystem

- **VIN:** The unregulated input voltage pin when supplying power via an external DC supply or battery pack (7 V–12 V).
- **5V:** The regulated +5.0 V power supply output derived from on-board linear voltage regulators (NCP1117 or LP2985). Used to power external sensors and integrated circuits.
- **3.3V:** A regulated +3.3 V supply generated on-board, delivering up to 50 mA for low-voltage sensor modules (e.g., ESP8266, XBee).
- **GND:** Ground reference pins (0 V).
- **IOREF:** Supplies the reference voltage at which the microcontroller I/O pins operate (5 V on standard UNO), allowing shield boards to adapt voltage levels.

2. Analog Input Subsystem (Pins A0 – A5)

The Arduino UNO features a 10-bit **Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC)** multiplexed across 6 channels (A0 to A5).

- **Resolution:** $2^{10} = 1024$ discrete digital quantization levels (0 to 1023).
- **Step Size (Quantization Voltage):**

$$\Delta V = \frac{V_{ref}}{1024} = \frac{5.0 \text{ V}}{1024} = 4.8828 \text{ mV/step} \quad (2.2)$$

- **Reading Function:** ‘analogRead(pin)’ returns an integer in the range [0, 1023].
- **Voltage Calculation Formula:**

$$V_{measured} = \text{ADC_Value} \times \left(\frac{V_{ref}}{1023} \right) \quad (2.3)$$

3. Digital Input/Output Subsystem (Pins 0 – 13)

Each digital pin can be programmed in software as an **INPUT**, **INPUT_PULLUP**, or **OUTPUT**.

- **UART Serial Pins:**
 - **Pin 0 (RX):** Receives asynchronous TTL serial data.
 - **Pin 1 (TX):** Transmits asynchronous TTL serial data.
- **External Hardware Interrupts:**
 - **Pin 2 (Interrupt 0 / INT0)** and **Pin 3 (Interrupt 1 / INT1)** can trigger immediate asynchronous ISR execution on ‘LOW’, ‘CHANGE’, ‘RISING’, or ‘FALLING’ signal transitions via ‘attachInterrupt()’.
- **Pulse Width Modulation (PWM) Subsystem (Pins 3, 5, 6, 9, 10, 11):** Simulates variable analog voltage output by rapidly switching digital pins between 0 V and 5 V at fixed carrier frequencies:
 - Pins 5 and 6: Carrier frequency $\approx 980 \text{ Hz}$ (governed by internal Timer 0).
 - Pins 3, 9, 10, 11: Carrier frequency $\approx 490 \text{ Hz}$ (governed by Timers 1 and 2).
 - Output resolution: 8-bit resolution (0 to 255) via ‘analogWrite(pin, value)’.

$$V_{avg} = V_{CC} \times \left(\frac{\text{PWM Value}}{255} \right) = 5.0 \text{ V} \times \text{Duty Cycle} \quad (2.4)$$

- **SPI Bus Subsystem:** Pin 10 (SS), Pin 11 (MOSI), Pin 12 (MISO), Pin 13 (SCK).
- **On-Board LED:** Connected directly to Pin 13.

Worked Example: ADC Voltage & PWM Calculation

An Arduino analog input pin A0 reads a value of 682 from an analog sensor.

1. Calculate the actual sensor voltage corresponding to this reading (assuming $V_{ref} = 5.0\text{ V}$).
2. What 8-bit PWM integer value must be passed to 'analogWrite(9, value)' to reproduce this exact average voltage across an LED?

Step 1: Given Data

- ADC Value = 682, Max ADC = 1023, $V_{ref} = 5.0\text{ V}$

Step 2: Calculate Measured Sensor Voltage

$$V_{measured} = \text{ADC Value} \times \frac{V_{ref}}{1023} = 682 \times \frac{5.0\text{ V}}{1023} \approx 3.333\text{ V}$$

Step 3: Calculate Required PWM Integer Value The average voltage output by PWM is:

$$V_{avg} = 5.0 \times \left(\frac{\text{PWM Value}}{255} \right) \Rightarrow \text{PWM Value} = 255 \times \left(\frac{V_{avg}}{5.0} \right)$$

Substitute $V_{avg} = 3.333\text{ V}$:

$$\text{PWM Value} = 255 \times \left(\frac{3.333}{5.0} \right) = 255 \times 0.6666 = 170$$

Final Answers:

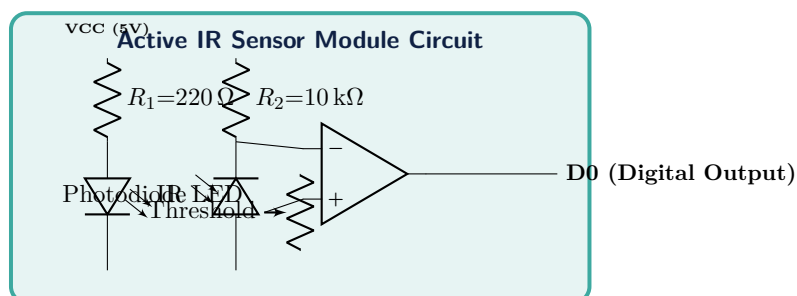
$$V_{measured} = 3.33\text{ V}, \quad \text{PWM Value} = \mathbf{170} \text{ (Duty Cycle } \approx \mathbf{66.7\%})$$

2.4 Sensors and Transducers

A **Transducer** is a physical device that converts energy from one physical form into another. A **Sensor** is an input transducer that measures a physical parameter (light, temperature, distance) and translates it into a measurable electrical signal (voltage, resistance, frequency).

2.4.1 Active Infrared (IR) Obstacle & Line Tracking Sensor

Active IR sensor modules detect object presence, proximity, or surface contrast based on infrared radiation reflection.



- **Subsystems:**

1. **IR Transmitter (IR LED):** Emits continuous infrared light at wavelength $\lambda \approx 940$ nm.
2. **IR Receiver (Photodiode):** Placed parallel to the transmitter, operating in reverse bias. When incident IR photons strike its junction, minority carrier photocurrent increases, causing its reverse resistance to drop sharply.
3. **Comparator (LM393 Op-Amp):** Compares the photodiode voltage against an adjustable reference voltage established by an on-board potentiometer.

- **Surface Albedo Principle (Line-Tracking Robots):**

- **White / Reflective Surface:** High albedo causes strong IR reflection. The photodiode receives high IR intensity $\Rightarrow$ resistance drops $\Rightarrow$ Comparator switches output to ****Logic LOW (0 V)****.
- **Black / Matte Surface:** Absorbs infrared radiation completely. Zero reflected IR reaches photodiode $\Rightarrow$ high resistance $\Rightarrow$ Comparator output remains ****Logic HIGH (5 V)****.

2.4.2 Light Dependent Resistor (LDR / Photoresistor)

An LDR is a passive photo-sensitive variable resistor constructed from semiconductor materials such as ****Cadmium Sulfide (CdS)**** or ****Cadmium Selenide (CdSe)****.

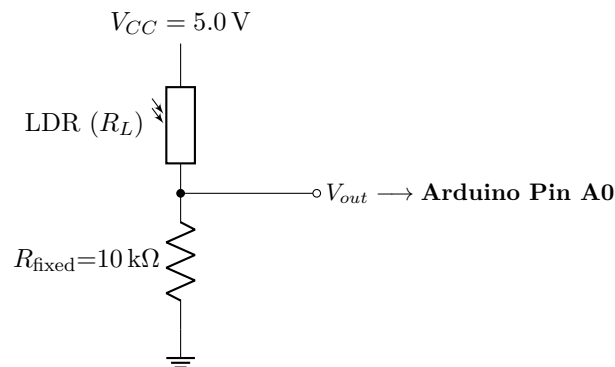


Figure 2.3: LDR Voltage Divider interfacing circuit with Arduino analog input.

- **Photoconductivity Physics:** In absolute darkness, valence electrons are locked within covalent bonds. The dark resistance is extremely high ($R_{dark} > 1 \text{ M}\Omega$). When incident photons with energy $h\nu \geq E_g$ strike the CdS track, valence electrons are excited across the bandgap into the conduction band, generating free electron-hole pairs. Consequently, electrical conductivity increases and resistance falls to a few hundred ohms ($R_{light} \approx 400 \Omega$).
- **Voltage Divider Analysis:** Because microcontrollers cannot measure resistance directly, the LDR is paired with a fixed precision resistor ($R_{fixed} = 10 \text{ k}\Omega$):

$$V_{out} = V_{CC} \left(\frac{R_{fixed}}{R_{LDR} + R_{fixed}} \right) \quad (2.5)$$

As light intensity increases $\Rightarrow R_{LDR}$ decreases $\Rightarrow V_{out}$ increases towards 5 V.

2.4.3 Ultrasonic Distance Sensor (HC-SR04)

The HC-SR04 rangefinder calculates physical distance to target objects by measuring acoustic ****Time-of-Flight (ToF)**** using ultrasonic pressure waves ($f = 40 \text{ kHz}$, beyond human hearing threshold).

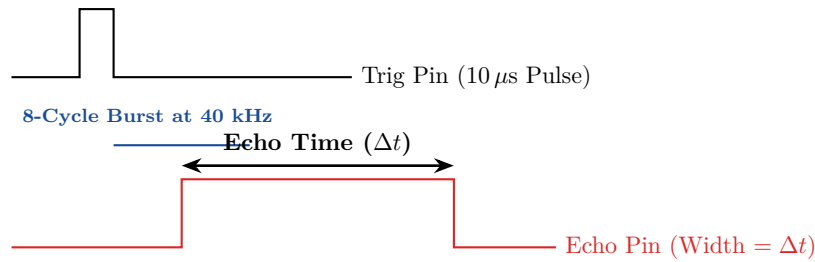


Figure 2.4: Operational timing protocol of the Ultrasonic Sensor HC-SR04.

- **Operational Protocol:**

1. The Arduino applies a 10 μs digital HIGH pulse to the **Trig** pin.
2. The sensor's on-board ultrasonic transducer emits an 8-cycle burst of acoustic waves at 40 kHz.
3. Simultaneously, the **Echo** pin transitions to logic HIGH (5 V).
4. The sound wave propagates through air at speed $v \approx 343 \text{ m} \cdot \text{s}^{-1}$, reflects off the obstacle, and returns to the receiving transducer.
5. Upon detecting the return echo, the sensor drives the **Echo** pin back to logic LOW.
6. The duration Δt for which the Echo pin remained HIGH equals the round-trip acoustic travel time.

- **Mathematical Derivation of Distance:**

$$\text{Total Distance Traveled} = v_{\text{sound}} \times \Delta t \quad (2.6)$$

Since the wave travels from sensor to target and back to sensor:

$$\text{Distance to Target } (D) = \frac{v_{\text{sound}} \times \Delta t}{2} \quad (2.7)$$

Using $v_{\text{sound}} = 343 \text{ m} \cdot \text{s}^{-1} = 0.0343 \text{ cm} \cdot \mu\text{s}^{-1}$:

$$D (\text{cm}) = \frac{\Delta t (\mu\text{s}) \times 0.0343}{2} = \frac{\Delta t (\mu\text{s})}{58.3} \quad (2.8)$$

2.4.4 Temperature and Humidity Sensors: DHT11 vs. DHT22

The DHT-series sensors integrate digital temperature and humidity measurement on a single silicon substrate with built-in ADC and single-wire communication protocol.

- **Sensing Mechanisms:**

- **Relative Humidity:** Measured using a capacitive humidity sensor consisting of a moisture-holding dielectric polymer sandwiched between two porous electrodes. When humidity changes, water vapor alters dielectric permittivity ϵ_r , directly varying capacitance $C = \frac{\epsilon_r \epsilon_0 A}{d}$.
- **Temperature:** Measured using a high-precision **NTC (Negative Temperature Coefficient) Thermistor**, whose resistance decreases non-linearly with increasing temperature.

- **Data Protocol:** Communicates via a single bidirectional digital data pin. Transmits a complete **40-bit data packet**:

$$40\text{-Bit Frame} = \text{Hum.Int}(8\text{b}) + \text{Hum.Dec}(8\text{b}) + \text{Temp.Int}(8\text{b}) + \text{Temp.Dec}(8\text{b}) + \text{Checksum}(8\text{b})$$

$$\text{Checksum Check: Byte 5} = (\text{Byte 1} + \text{Byte 2} + \text{Byte 3} + \text{Byte 4}) \pmod{256}$$

Specification Parameter	DHT11 Sensor	DHT22 Sensor (AM2302)
Operating Voltage	3.3 V to 5.5 V DC	3.3 V to 6.0 V DC
Humidity Sensing Range	20% to 90% RH	0% to 100% RH
Humidity Accuracy	±5% RH	±2% RH
Temperature Range	0 °C to 50 °C	−40 °C to 80 °C
Temperature Accuracy	±2.0 °C	±0.5 °C
Sampling Rate	1 Hz (1 reading/second)	0.5 Hz (1 reading/2 seconds)
Cost & Target Application	Low cost / Educational kits	Precision industrial / Environmental monitoring

Table 2.2: Comprehensive Comparison between DHT11 and DHT22 Sensors.

2.5 Summary & Chapter Review

Quick Revision: Unit 2 Key Takeaways

- **Signals:** Analog is continuous with infinite resolution; Digital is discrete binary (HIGH/LOW) with high noise immunity.
- **ATmega328P Specs:** 8-bit AVR RISC, 16 MHz, 5V, 32KB Flash, 2KB SRAM, 1KB EEPROM.
- **Arduino ADC:** 10-bit resolution (0–1023), step size = 4.88 mV.
- **Arduino PWM:** 8-bit resolution (0–255) on pins 3, 5, 6, 9, 10, 11.
- **IR Sensor:** Detects reflection; white reflects (Low output), black absorbs (High output).
- **LDR:** CdS photoconductivity; higher illuminance reduces resistance.
- **HC-SR04:** Distance = $\frac{t_{\text{echo}} \times 0.0343}{2} = \frac{t_{\text{echo}}(\mu\text{s})}{58.3}$.
- **DHT11/22:** 40-bit digital stream measuring humidity (capacitive) and temperature (NTC thermistor).

Number Systems, Binary Codes, Logic Gates & Karnaugh Maps

3.1 Introduction

Digital computing and communication systems process information exclusively in discrete mathematical abstractions. This chapter establishes the structural foundations of digital systems: multi-base number representations, error-resilient binary codes, 2's complement arithmetic, Boolean algebra postulates, universal gate implementations, and visual minimization using Karnaugh Maps (K-Maps).

3.2 Positional Number Systems & Base Conversions

3.2.1 Positional Notation

In a positional number system with base (or radix) r , any real number N is expressed as an ordered sequence of digits:

$$N = (d_{n-1}d_{n-2} \cdots d_1d_0.d_{-1}d_{-2} \cdots d_{-m})_r \quad (3.1)$$

The numerical value is evaluated by the weighted polynomial expansion:

Positional Weighted Sum Formula

$$N = \sum_{i=-m}^{n-1} d_i \cdot r^i = d_{n-1}r^{n-1} + \cdots + d_1r^1 + d_0r^0 + d_{-1}r^{-1} + \cdots + d_{-m}r^{-m} \quad (3.2)$$

System Name	Base (r)	Permissible Digits	Example Expression
Binary	2	0, 1	$(11010.11)_2$
Octal	8	0, 1, 2, 3, 4, 5, 6, 7	$(754.2)_8$
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9	$(254.625)_{10}$
Hexadecimal	16	0–9, A–F (where A=10, ..., F=15)	$(12F8A.C)_{16}$

Table 3.1: Primary positional number systems used in computing architecture.

3.2.2 Multi-Base Conversion Algorithms

1. Decimal to Base- r Conversion

- **Integer Part (Successive Division):** Divide the integer repeatedly by the destination radix r . Record the integer quotient and remainder at each step. Terminate when quotient reaches 0. The base- r integer is obtained by reading remainders from ****bottom to top** (MSB to LSB)**.
- **Fractional Part (Successive Multiplication):** Multiply the fractional component repeatedly by r . Record the resulting integer digit and retain the fractional product. Terminate when fraction becomes zero or desired precision is achieved. The base- r fraction is read from ****top to bottom****.

Worked Example: Decimal to Binary Conversion

Convert $(53.6875)_{10}$ into its binary equivalent.

Integer Conversion (53 by 2):

$$\begin{aligned}
 53 \div 2 &= 26 \quad \text{rem } 1 \quad (\text{LSB}) \\
 26 \div 2 &= 13 \quad \text{rem } 0 \\
 13 \div 2 &= 6 \quad \text{rem } 1 \\
 6 \div 2 &= 3 \quad \text{rem } 0 \\
 3 \div 2 &= 1 \quad \text{rem } 1 \\
 1 \div 2 &= 0 \quad \text{rem } 1 \quad (\text{MSB}) \implies (53)_{10} = (110101)_2
 \end{aligned}$$

Fractional Conversion (0.6875 by 2):

$$\begin{aligned}
 0.6875 \times 2 &= 1.3750 \implies \text{Integer: } 1 \\
 0.3750 \times 2 &= 0.7500 \implies \text{Integer: } 0 \\
 0.7500 \times 2 &= 1.5000 \implies \text{Integer: } 1 \\
 0.5000 \times 2 &= 1.0000 \implies \text{Integer: } 1 \implies (0.6875)_{10} = (.1011)_2
 \end{aligned}$$

Final Answer: $(53.6875)_{10} = (110101.1011)_2$

2. Base- r to Decimal Conversion

Evaluate the weighted polynomial sum $\sum d_i \cdot r^i$.

3. Binary $\leftrightarrow$ Octal / Hexadecimal Conversions

Because $8 = 2^3$ and $16 = 2^4$, conversions between these bases do not require intermediate decimal arithmetic:

- **Binary to Octal:** Partition bits into groups of **3 bits** starting from the radix point (leftward for integers, rightward for fractions, padding with zeros if needed).
- **Binary to Hexadecimal:** Partition bits into groups of **4 bits** starting from the radix point.

3.3 Binary Codes

3.3.1 Weighted vs. Non-Weighted Codes

- **Weighted Codes:** Each bit position has a strictly assigned numerical weight (e.g., Natural Binary, BCD 8421, 2421 code). The value is $\sum b_i W_i$.
- **Non-Weighted Codes:** Bit positions do not have fixed weights; code assignment is based on specific mathematical properties (e.g., Excess-3, Gray code).

3.3.2 Binary Coded Decimal (BCD / 8421 Code)

BCD represents each decimal digit (0 through 9) individually using a 4-bit binary group (8–4–2–1 weighting).

- **Valid BCD Codes:** 0000_2 to 1001_2 (0 to 9).
- **Invalid BCD States (6 forbidden states):** $1010_2, 1011_2, 1100_2, 1101_2, 1110_2, 1111_2$ (10 to 15).
- **BCD Addition Rule:** If binary sum of any digit group exceeds 9 (1001_2) or produces an output carry, add 0110_2 (6_{10}) to skip the six invalid states.

3.3.3 Excess-3 Code (XS-3)

Excess-3 is an unweighted code obtained by adding 3 (0011_2) to each BCD code group.

Definition: Self-Complementing Property

A code is self-complementing if the 9's complement of a decimal digit is obtained directly by inverting all bits (1's complement) of its code word. Excess-3 is self-complementing.

For example:

- Decimal 2 $\implies 2 + 3 = 5 \implies$ XS-3: 0101_2 .
- 1's complement of $0101_2 = 1010_2 \implies 1010_2$ in XS-3 is $10 - 3 = 7_{10}$.
- Note that 7_{10} is exactly the 9's complement of 2_{10} ($9 - 2 = 7$).

3.3.4 Gray Code (Reflected Binary Code)

Gray Code is an unweighted ****Unit-Distance Code****: exactly ****one bit**** changes between any two consecutive numbers.

Exam Tip: Why Gray Code Matters in Engineering

When mechanical shaft encoders or asynchronous counters transition across states (e.g., $011_2 \rightarrow 100_2$), slight timing skews cause multiple bits to change at slightly different instants, creating temporary false intermediate glitch states ($011 \rightarrow 010 \rightarrow 000 \rightarrow 100$). Gray code guarantees that only one bit changes per step, eliminating switching glitches completely.

Binary $\longleftrightarrow$ Gray Code Conversion Algorithms

Binary to Gray Code:

$$G_{n-1} = B_{n-1} \quad (\text{MSB}), \quad G_i = B_{i+1} \oplus B_i \quad \text{for } i = n-2, \dots, 0 \quad (3.3)$$

Gray Code to Binary:

$$B_{n-1} = G_{n-1} \quad (\text{MSB}), \quad B_i = B_{i+1} \oplus G_i \quad \text{for } i = n-2, \dots, 0 \quad (3.4)$$

3.4 Complements & Fixed-Point Binary Arithmetic

3.4.1 Radix and Diminished Radix Complements

For a base- r system with n digits:

- **Radix (r 's) Complement:** $r^n - N$
- **Diminished Radix ($(r-1)$'s) Complement:** $(r^n - 1) - N$
- **Relationship:** r 's complement = $(r-1)$'s complement + 1

In the Binary System ($r = 2$):

- **1's Complement:** Invert every bit individually ($0 \leftrightarrow 1$).
- **2's Complement:** 1's Complement + 1.

3.4.2 2's Complement Subtraction ($A - B$)

Digital arithmetic logic units (ALUs) perform subtraction using addition hardware via 2's complement representation.

2's Complement Subtraction Procedure

To compute $A - B$:

1. Express both numbers using an identical bit-width n .
2. Find the 2's complement of subtrahend B : $B_{2's} = \overline{B} + 1$.
3. Add: $Sum = A + B_{2's}$.
4. **Case 1: End-Around Carry Generated ($C_{out} = 1$)**
 - The result is **Positive**.
 - Discard the end-around carry. The remaining bits form the true binary magnitude.
5. **Case 2: No End-Around Carry ($C_{out} = 0$)**
 - The result is **Negative**.
 - The magnitude is obtained by taking the 2's complement of the Sum , prefixed with a negative sign.

Worked Example: 2's Complement Subtraction Calculations

Perform the following subtractions using 8-bit 2's complement arithmetic:

1. $(78)_{10} - (35)_{10}$
2. $(35)_{10} - (78)_{10}$

Binary Representations (8-bit):

- $(78)_{10} = 01001110_2$
- $(35)_{10} = 00100011_2$

Part 1: $(78)_{10} - (35)_{10}$

$$1\text{'s comp of } 35 = 11011100_2$$

$$2\text{'s comp of } 35 = 11011100_2 + 1 = 11011101_2$$

$$\text{Add: } 78 + (-35) = 01001110_2 + 11011101_2 = \mathbf{1\ 00101011_2}$$

End-around carry is 1 $\Rightarrow$ Result is Positive. Discard carry:

$$00101011_2 = 32 + 8 + 2 + 1 = (+43)_{10} \quad \checkmark$$

Part 2: $(35)_{10} - (78)_{10}$

$$1\text{'s comp of } 78 = 10110001_2$$

$$2\text{'s comp of } 78 = 10110001_2 + 1 = 10110010_2$$

$$\text{Add: } 35 + (-78) = 00100011_2 + 10110010_2 = 11010101_2$$

No carry generated ($C_{out} = 0$) $\Rightarrow$ Result is Negative.

$$\text{Magnitude} = 2\text{'s comp of } 11010101_2 = \overline{11010101} + 1 = 00101010_2 + 1 = 00101011_2 = (43)_{10}$$

$$\text{Result} = -(43)_{10} \quad \checkmark$$

3.5 Logic Gates & Boolean Algebra

3.5.1 Standard Logic Gates

3.5.2 Universality of NAND and NOR Gates

Any Boolean function can be implemented using exclusively NAND gates or exclusively NOR gates.

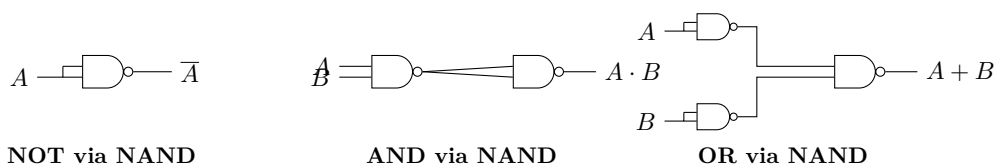


Figure 3.1: Realization of basic gates using exclusively universal NAND gates.

3.5.3 Boolean Algebra Postulates, Laws, and De Morgan's Theorems

3.6 Canonical Forms & Karnaugh Maps (K-Maps)

Gate Name	Symbol	Boolean Expression	Truth Table ($A, B \rightarrow Y$)	Summary Operation
NOT (Inverter)	Triangle+bubble	$Y = \overline{A}$	$0 \rightarrow 1, 1 \rightarrow 0$	Inverts binary state.
AND	D-shape	$Y = A \cdot B$	00: 0, 01: 0 10: 0, 11: 1	Output 1 only if all inputs are 1.
OR	Curved input	$Y = A + B$	00: 0, 01: 1 10: 1, 11: 1	Output 1 if any input is 1.
NAND	AND+bubble	$Y = \overline{A \cdot B}$	00: 1, 01: 1 10: 1, 11: 0	Universal gate; inverted AND.
NOR	OR+bubble	$Y = \overline{A + B}$	00: 1, 01: 0 10: 0, 11: 0	Universal gate; inverted OR.
XOR	Double curve	$Y = A \oplus B = \overline{A}B + A\overline{B}$	00: 0, 01: 1 10: 1, 11: 0	Output 1 if inputs are unequal.
XNOR	XOR+bubble	$Y = \overline{A \oplus B} = AB + \overline{A}\overline{B}$	00: 1, 01: 0 10: 0, 11: 1	Output 1 if inputs are identical.

Table 3.2: The seven primary digital logic gates.

Law / Property	OR Form (Dual 1)	AND Form (Dual 2)
Identity Law	$A + 0 = A$	$A \cdot 1 = A$
Null / Domination Law	$A + 1 = 1$	$A \cdot 0 = 0$
Idempotent Law	$A + A = A$	$A \cdot A = A$
Involution Law	$\overline{\overline{A}} = A$	$\overline{\overline{A}} = A$
Complementarity Law	$A + \overline{A} = 1$	$A \cdot \overline{A} = 0$
Commutative Law	$A + B = B + A$	$A \cdot B = B \cdot A$
Associative Law	$A + (B + C) = (A + B) + C$	$A \cdot (B \cdot C) = (A \cdot B) \cdot C$
Distributive Law	$A \cdot (B + C) = AB + AC$	$A + BC = (A + B)(A + C)$
Absorption Law	$A + AB = A$	$A(A + B) = A$
Redundant Literal Law	$A + \overline{A}B = A + B$	$A(\overline{A} + B) = AB$
De Morgan's Theorem 1	$\overline{A \cdot B} = \overline{A} + \overline{B}$	$\overline{A_1 \cdot A_2 \cdots A_n} = \sum \overline{A_i}$
De Morgan's Theorem 2	$\overline{A + B} = \overline{A} \cdot \overline{B}$	$\overline{A_1 + A_2 + \cdots + A_n} = \prod \overline{A_i}$

Table 3.3: Complete Boolean algebra postulates, identities, and dual theorems.

3.6.1 Canonical SOP (Minterms) vs. Canonical POS (Maxterms)

- **Minterm (m_i):** A standard product term containing all system variables in either true or complemented form. A minterm equals 1 for exactly one input combination. In minterms, a variable is uncomplemented (A) if its value is 1, and complemented ($\overline{A}$) if its value is 0.
- **Maxterm (M_i):** A standard sum term containing all system variables. A maxterm equals 0 for exactly one input combination. In maxterms, a variable is uncomplemented (A) if its value is 0, and complemented ($\overline{A}$) if its value is 1.
- **Duality Relationship:**

$$m_i = \overline{M_i} \iff M_i = \overline{m_i} \quad (3.5)$$

$$F(A, B, C) = \sum m(1, 3, 5, 7) = \prod M(0, 2, 4, 6) \quad (3.6)$$

3.6.2 K-Map Structure and Cell Gray Code Adjacency

A Karnaugh Map arranges truth table outputs onto a multi-dimensional grid where adjacent cells differ by exactly **one binary variable** (Gray code ordering: 00, 01, 11, 10).

AB \ CD	CD			
	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

Figure 3.2: Standard 4-variable Karnaugh Map cell coordinates and decimal minterm designations.

3.6.3 K-Map Minimization Rules

1. **Group Size:** Groups must contain 2^k adjacent cells (1, 2, 4, 8, 16).
2. **Variable Elimination Rule:**
 - A group of 2 cells (Pair) eliminates 1 variable.

- A group of 4 cells (Quad) eliminates 2 variables.
 - A group of 8 cells (Octet) eliminates 3 variables.
 - A group of 16 cells eliminates all variables ($F = 1$).
3. **Toroidal Wrap-Around:** The top and bottom rows are adjacent; the leftmost and rightmost columns are adjacent.
 4. **Corner Quad:** Cells (m_0, m_2, m_8, m_{10}) form a valid 4-cell quad minimizing to $\overline{B}\overline{D}$.
 5. **Don't Care Conditions (d or X):** Set to 1 if they help form a larger prime implicant group; otherwise, set to 0.

Worked Example: 4-Variable K-Map Minimization with Don't Cares

Minimize the Boolean function:

$$F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 14, 15) + d(3, 11)$$

Step 1: Plotting on the K-Map

- 1s at: 0, 2, 5, 7, 8, 10, 14, 15
- Xs at: 3, 11

Step 2: Grouping Strategy

1. Corner Quad (m_0, m_2, m_8, m_{10}):

- Rows 00 and 10 $\Rightarrow B$ changes ($0 \rightarrow 1$), $A = 0$ and $A = 1$, so $B = 0$ is constant $\Rightarrow \overline{B}$.
- Columns 00 and 10 $\Rightarrow C$ changes, $D = 0$ is constant $\Rightarrow \overline{D}$.
- **Term 1:** $\overline{B}\overline{D}$

2. Center Quad ($m_5, m_7, m_{13}(\text{empty}), m_{15}?$ **No:** m_5, m_7 with $d_3, m_{11}(\text{empty})?$):

Look at cells $m_5(0101), m_7(0111)$ along with $d_3(0011)$ and $m_{11}(0110)$? Pair m_5, m_7 : Row 01 $\Rightarrow \overline{A}B$, Col 01, 11 $\Rightarrow D$ constant. Term: $\overline{A}BD$. Can we combine m_5, m_7 with d_3, d_{11} ? No, d_3, d_{11} are in col 11. Look at Quad in column 11: (d_3, m_7, m_{15}, d_{11})!

- Covers m_7, m_{15} along with don't cares d_3, d_{11} .
- All 4 rows covered $\Rightarrow A, B$ eliminated.
- Column 11 $\Rightarrow CD$.
- **Term 2:** CD

3. Remaining 1s:

- $m_5(0101)$: Group with $m_7(0111) \Rightarrow$ Pair (m_5, m_7) $\Rightarrow \overline{A}BD$.
- $m_{14}(1110)$: Group with $m_{15}(1111) \Rightarrow$ Pair (m_{14}, m_{15}) $\Rightarrow ABC$.
- Alternatively, check if m_{14} can group with $m_{10}(1010)$: Pair (m_{10}, m_{14}) $\Rightarrow ACD$.

Final Simplified Equation:

$$F(A, B, C, D) = \overline{B}\overline{D} + CD + \overline{A}BD + ABC$$

3.7 Summary & Chapter Review

Quick Revision: Unit 3 Key Formula & Takeaway Sheet

- **Base Conversion:** Successive division/multiplication for decimal $\leftrightarrow$ base r ; 3-bit and 4-bit grouping for Octal/Hex $\leftrightarrow$ Binary.
- **BCD:** 8421 code; values > 9 invalid; add 0110_2 to correct.
- **Excess-3:** BCD + 0011_2 ; self-complementing.
- **Gray Code:** $G_i = B_{i+1} \oplus B_i$; unit distance property eliminates glitches.
- **2's Complement Subtraction:** $A - B = A + (\bar{B} + 1)$. Carry = 1 $\implies$ positive (discard carry); Carry = 0 $\implies$ negative (take 2's comp of sum).
- **Universal Gates:** NAND and NOR can synthesize any logic network.
- **De Morgan's Laws:** $\overline{A \cdot B} = \bar{A} + \bar{B}$; $\overline{A + B} = \bar{A} \cdot \bar{B}$.
- **K-Map Simplification:** Gray code grid adjacency; Pair (1 var), Quad (2 vars), Octet (3 vars); Corners $(0, 2, 8, 10) = \bar{B}\bar{D}$.

Introduction to Combinational Logic Circuits

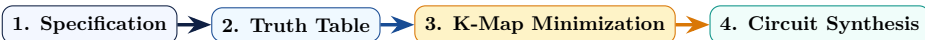
4.1 Introduction & Design Principles

Combinational logic circuits are the fundamental building blocks of digital processing architectures. In a combinational circuit, the outputs at any instant of time depend **strictly** on the present combination of input variables, with no influence from past inputs or stored states.

Definition: Combinational Logic Circuit

A network of interconnected digital logic gates whose outputs at any given time are determined solely by the instantaneous values of its inputs, without feedback paths or internal memory elements.

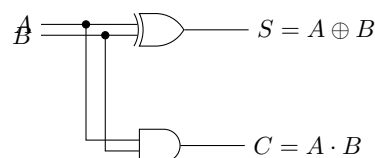
The Standard Combinational Design Flow:



4.2 Arithmetic Adders

4.2.1 Half Adder

A Half Adder is an arithmetic circuit that adds two single-bit binary inputs (A and B) and generates two output bits: **Sum (S)** and **Carry (C)**.



Input A	Input B	Carry (C)	Sum (S)	Arithmetic Addition
0	0	0	0	$0 + 0 = 00_2$
0	1	0	1	$0 + 1 = 01_2$
1	0	0	1	$1 + 0 = 01_2$
1	1	1	0	$1 + 1 = 10_2$

Table 4.1: Truth table for a Binary Half Adder.

Half Adder Boolean Expressions

$$S = A \oplus B = \overline{A}B + A\overline{B}, \quad C = A \cdot B \quad (4.1)$$

4.2.2 Full Adder

A Half Adder cannot accept a carry generated from a lower significant column. A **Full Adder** adds three 1-bit inputs: A , B , and an incoming carry C_{in} , producing Sum (S) and Carry-Out (C_{out}).

Input A	Input B	Input C_{in}	Carry Out (C_{out})	Sum (S)	Decimal Value
0	0	0	0	0	$0 + 0 + 0 = 0$
0	0	1	0	1	$0 + 0 + 1 = 1$
0	1	0	0	1	$0 + 1 + 0 = 1$
0	1	1	1	0	$0 + 1 + 1 = 2$
1	0	0	0	1	$1 + 0 + 0 = 1$
1	0	1	1	0	$1 + 0 + 1 = 2$
1	1	0	1	0	$1 + 1 + 0 = 2$
1	1	1	1	1	$1 + 1 + 1 = 3$

Table 4.2: Truth table for a Binary Full Adder.

Full Adder Boolean Equations

$$S = A \oplus B \oplus C_{in} \quad (4.2)$$

$$C_{out} = AB + BC_{in} + AC_{in} = AB + C_{in}(A \oplus B) \quad (4.3)$$

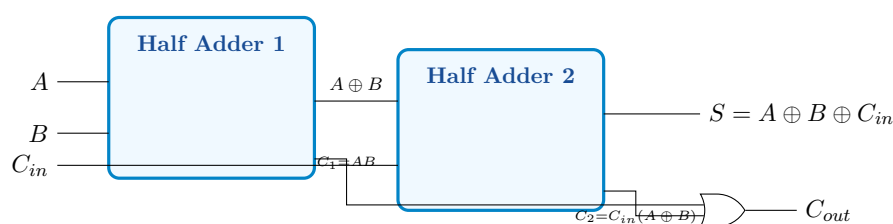


Figure 4.1: Full Adder synthesized using Two Half Adders and One OR Gate.

4.3 Arithmetic Subtractors

4.3.1 Half Subtractor

Subtracts a 1-bit subtrahend (B) from a 1-bit minuend (A), generating **Difference (D)** and **Borrow-Out (B_{out})**.

Minuend (A)	Subtrahend (B)	Borrow (B_{out})	Difference (D)	Arithmetic Operation ($A - B$)
0	0	0	0	$0 - 0 = 0$
0	1	1	1	$0 - 1 = -1 \implies$ Borrow 1 $\rightarrow$ Diff 1
1	0	0	1	$1 - 0 = 1$
1	1	0	0	$1 - 1 = 0$

Table 4.3: Truth table for a Binary Half Subtractor.

Half Subtractor Boolean Equations

$$D = A \oplus B, \quad B_{out} = \bar{A} \cdot B \quad (4.4)$$

4.3.2 Full Subtractor

Performs binary subtraction involving three bits: minuend (A), subtrahend (B), and borrow-in (B_{in}).

Full Subtractor Boolean Equations

$$D = A \oplus B \oplus B_{in} \quad (4.5)$$

$$B_{out} = \bar{A}B + \bar{A}B_{in} + BB_{in} = \bar{A}B + B_{in}(\bar{A} \oplus B) \quad (4.6)$$

A Full Subtractor can be realized by cascading **two Half Subtractors and one OR gate**.

4.4 Multiplexers (MUX / Data Selectors)**4.4.1 Theory of Operation**

A Multiplexer (MUX) is a combinational circuit that selects binary information from one of 2^n input lines and directs it to a single output line based on the binary state of n selection lines.

Multiplexer Fundamental Dimensioning

$$N_{inputs} = 2^n \iff n = \log_2(N_{inputs}) \quad (4.7)$$

Where N_{inputs} is the number of data input lines and n is the number of select lines.

4.4.2 4:1 Multiplexer

- **Data Inputs:** I_0, I_1, I_2, I_3 ($2^2 = 4 \implies n = 2$)
- **Select Lines:** S_1, S_0 (S_1 is MSB, S_0 is LSB)

- **Output: Y**

Select S_1	Select S_0	Output Y	Selected Channel
0	0	I_0	Channel 0 Selected
0	1	I_1	Channel 1 Selected
1	0	I_2	Channel 2 Selected
1	1	I_3	Channel 3 Selected

Table 4.4: Function Table for a 4:1 Multiplexer.

4:1 Multiplexer Output Equation

$$Y = \overline{S_1} \overline{S_0} I_0 + \overline{S_1} S_0 I_1 + S_1 \overline{S_0} I_2 + S_1 S_0 I_3 \quad (4.8)$$

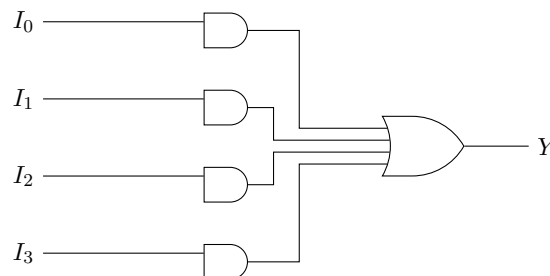


Figure 4.2: Gate-level logic implementation of a 4-to-1 Multiplexer.

Worked Example: Boolean Function Synthesis using an 8:1 Multiplexer

Implement the 3-variable logic function $F(A, B, C) = \sum m(1, 3, 4, 6, 7)$ using an 8:1 Multiplexer.

Step 1: Determine Dimensions Number of variables $n = 3 \implies 2^3 = 8$ data inputs (I_0 to I_7) and 3 select lines ($S_2 = A, S_1 = B, S_0 = C$).

Step 2: Map Minterms to Multiplexer Data Inputs

- Connect minterm indices present in F (1, 3, 4, 6, 7) to logic HIGH (1/ V_{CC}).
- Connect remaining minterm indices (0, 2, 5) to logic LOW (0/GND).

Step 3: Multiplexer Connections

$$I_1 = I_3 = I_4 = I_6 = I_7 = 1 \text{ (5V)}, \quad I_0 = I_2 = I_5 = 0 \text{ (GND)}$$

$$S_2 = A, \quad S_1 = B, \quad S_0 = C$$

4.5 De-Multiplexers (DEMUX / Data Distributors)

4.5.1 Theory of Operation

A De-multiplexer receives data on a single input line and steers it to one of 2^n output lines depending on the values of n selection lines.

1:4 De-multiplexer Equations

$$Y_0 = \overline{S_1} \overline{S_0} \cdot D_{in} \quad (4.9)$$

$$Y_1 = \overline{S_1} S_0 \cdot D_{in} \quad (4.10)$$

$$Y_2 = S_1 \overline{S_0} \cdot D_{in} \quad (4.11)$$

$$Y_3 = S_1 S_0 \cdot D_{in} \quad (4.12)$$

4.6 Decoders and Encoders

4.6.1 Binary Decoders (2-to-4 and 3-to-8 Lines)

A Decoder is a combinational circuit that converts an n -bit binary code into up to 2^n unique output lines.

- ****Active-High Outputs:**** The selected line outputs logic **1**; all non-selected lines output logic **0**.
- ****Active-Low Outputs (e.g., 74138 Decoder):**** The selected line outputs logic **0**; all non-selected lines output logic **1**.

Enable (E)	Input A	Input B	Y_3	Y_2	Y_1	Y_0
0	X	X	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

Table 4.5: Truth table for a 2-to-4 Line Decoder with Active-High outputs.

4.6.2 Binary Encoders & Priority Encoders

An Encoder performs the inverse operation of a decoder: it accepts 2^n (or fewer) input lines and generates an n -bit binary code.

8-to-3 Octal-to-Binary Encoder

- **Inputs:** $D_0, D_1, D_2, D_3, D_4, D_5, D_6, D_7$ (only one input is active at any time).
- **Outputs:** X, Y, Z (X is MSB, Z is LSB).

8-to-3 Encoder Output Equations

$$Z = D_1 + D_3 + D_5 + D_7 \quad (4.13)$$

$$Y = D_2 + D_3 + D_6 + D_7 \quad (4.14)$$

$$X = D_4 + D_5 + D_6 + D_7 \quad (4.15)$$

Common Pitfall: Priority Encoder Requirement

Standard encoders produce invalid outputs when more than one input is active simultaneously. A **Priority Encoder** establishes an explicit input priority (e.g., $D_7 > D_6 > \dots > D_0$). If both D_2 and D_6 are active, the priority encoder exclusively outputs code 110_2 (corresponding to D_6). It also provides a **Valid Output (V)** flag to differentiate input $D_0 = 1$ from an all-zero inactive condition.

4.7 Digital Magnitude Comparators

4.7.1 1-Bit Magnitude Comparator

Compares two 1-bit inputs A and B , generating three mutually exclusive outputs:

1-Bit Magnitude Comparator Equations

$$(A > B) = A \cdot \overline{B} \quad (4.16)$$

$$(A = B) = A \odot B = \overline{A \oplus B} = AB + \overline{A} \overline{B} \quad (4.17)$$

$$(A < B) = \overline{A} \cdot B \quad (4.18)$$

4.7.2 2-Bit Magnitude Comparator

Compares two 2-bit words: $A = (A_1 A_0)_2$ and $B = (B_1 B_0)_2$.

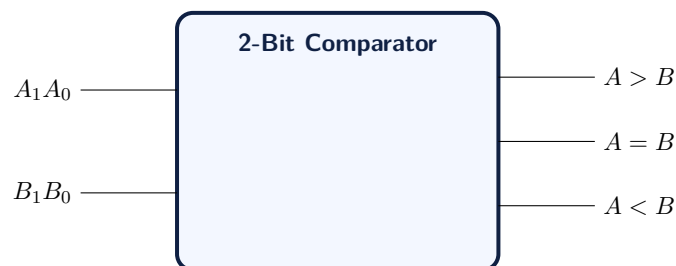


Figure 4.3: Block representation of a 2-bit digital magnitude comparator.

2-Bit Magnitude Comparator Equations

Let equality flags for individual bit positions be:

$$x_1 = \overline{A_1 \oplus B_1} = A_1 B_1 + \overline{A_1} \overline{B_1}, \quad x_0 = \overline{A_0 \oplus B_0} = A_0 B_0 + \overline{A_0} \overline{B_0} \quad (4.19)$$

Then the comparator outputs are synthesized as:

$$(A = B) = x_1 \cdot x_0 \quad (4.20)$$

$$(A > B) = A_1 \overline{B_1} + x_1 (A_0 \overline{B_0}) \quad (4.21)$$

$$(A < B) = \overline{A_1} B_1 + x_1 (\overline{A_0} B_0) \quad (4.22)$$

4.8 Summary & Chapter Review

Quick Revision: Unit 4 Essential Review Sheet

- **Combinational Circuit:** Memoryless; output depends strictly on instantaneous inputs.
- **Half Adder:** $S = A \oplus B$, $C = AB$.
- **Full Adder:** $S = A \oplus B \oplus C_{in}$, $C_{out} = AB + C_{in}(A \oplus B)$. Constructed with 2 Half Adders and 1 OR gate.
- **Half Subtractor:** $D = A \oplus B$, $B_{out} = \bar{A}B$.
- **Full Subtractor:** $D = A \oplus B \oplus B_{in}$, $B_{out} = \bar{A}B + B_{in}(\overline{A \oplus B})$. Constructed with 2 Half Subtractors and 1 OR gate.
- **Multiplexer (MUX):** 2^n data inputs, n select lines, 1 output. Acts as a universal logic synthesizer.
- **De-multiplexer (DEMUX):** 1 data input, n select lines, 2^n outputs.
- **Decoder:** n inputs $\rightarrow 2^n$ outputs. Active-High vs Active-Low (74138).
- **Encoder:** 2^n inputs $\rightarrow n$ outputs. Priority encoders resolve multi-input conflicts.
- **2-Bit Comparator:** Evaluates $A > B$, $A = B$, $A < B$ using bitwise equality terms x_1, x_0 .

Introduction to Sequential Logic Circuits

5.1 Introduction & Theoretical Overview

While combinational logic evaluates expressions instantaneously from current inputs, **sequential logic circuits** incorporate memory elements and feedback loops. The output of a sequential circuit is a function of both the **current external inputs** and the **present internal state** (stored historical information).

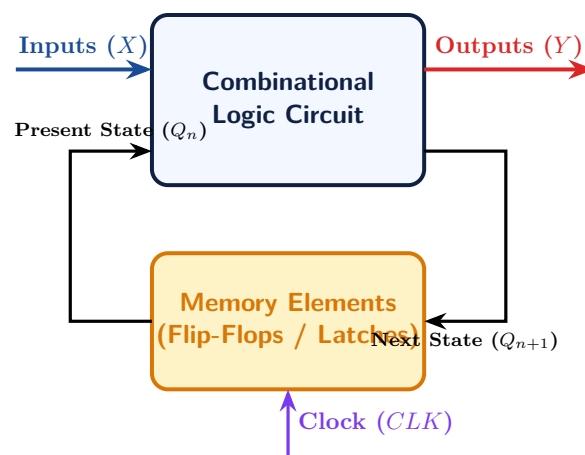


Figure 5.1: Canonical architectural model of a Synchronous Sequential Logic Circuit.

5.2 Bistable Latches

A **Latch** is a level-sensitive bistable multivibrator. It is asynchronous: outputs respond directly to input levels whenever the enable line is active.

5.2.1 Cross-Coupled NOR SR Latch (Active-High)

Constructed from two cross-coupled NOR gates with inputs S (Set) and R (Reset).

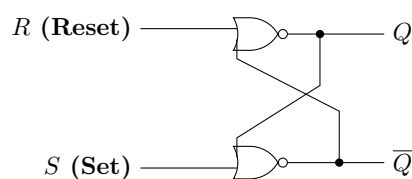


Figure 5.2: Logic gate implementation of an Active-High Cross-Coupled NOR SR Latch.

Set (S)	Reset (R)	Next State Q_{n+1}	$\overline{Q}_{n+1}$	Operating State / Mode
0	0	Q_n	$\overline{Q}_n$	Hold / No Change (Memory Mode)
0	1	0	1	Reset State ($Q \rightarrow 0$)
1	0	1	0	Set State ($Q \rightarrow 1$)
1	1	0	0	Invalid / Forbidden State ($Q = \overline{Q} = 0$)

Table 5.1: Operational Truth Table for Active-High NOR SR Latch.

5.2.2 Cross-Coupled NAND $\overline{S}\overline{R}$ Latch (Active-Low)

Constructed from two cross-coupled NAND gates. The inputs are labeled $\overline{S}$ and $\overline{R}$ because a logic 0 activates the respective Set or Reset function.

$\overline{S}$	$\overline{R}$	Next State Q_{n+1}	$\overline{Q}_{n+1}$	Operating State / Mode
0	0	1	1	Invalid / Forbidden State ($Q = \overline{Q} = 1$)
0	1	1	0	Set State ($Q \rightarrow 1$)
1	0	0	1	Reset State ($Q \rightarrow 0$)
1	1	Q_n	$\overline{Q}_n$	Hold / No Change (Memory Mode)

Table 5.2: Operational Truth Table for Active-Low NAND $\overline{S}\overline{R}$ Latch.

5.2.3 Gated D Latch (Transparent Data Latch)

The Gated D Latch eliminates the invalid/forbidden state of the SR latch by connecting an inverter between the Set and Reset inputs ($R = \overline{S} = \overline{D}$), and gating the inputs with an Enable (E) line.

- **When Enable $E = 1$ (Transparent Mode):** The output Q continuously follows the input D ($Q = D$).
- **When Enable $E = 0$ (Latching Mode):** The inputs are isolated, and the latch stores and holds its previous state Q_n indefinitely.

5.3 Edge-Triggered Flip-Flops

Flip-flops are synchronous bistable storage elements that sample their inputs and transition their output states ****only on a designated clock edge**** (Positive/Rising edge $\uparrow$ or Negative/Falling edge $\downarrow$).

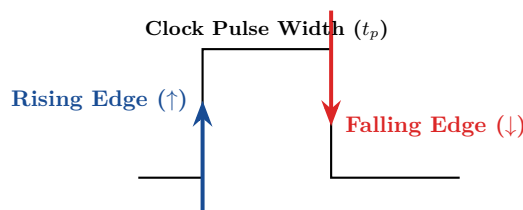


Figure 5.3: Clock pulse signal anatomy showing rising (positive) and falling (negative) trigger edges.

5.3.1 The Four Foundational Flip-Flops

1. Clocked SR Flip-Flop

- **Characteristic Equation:** $Q_{n+1} = S + \overline{R}Q_n$ (Valid for $SR = 0$)
- **Operation:** $S = 0, R = 0 \implies \text{Hold}$; $S = 0, R = 1 \implies 0$; $S = 1, R = 0 \implies 1$; $S = 1, R = 1 \implies \text{Invalid}$.

2. Clocked JK Flip-Flop

The JK flip-flop eliminates the invalid state of the SR flip-flop by feeding back the output states (Q and $\overline{Q}$) to the steering input gates.

- **Characteristic Equation:**

$$Q_{n+1} = J\overline{Q}_n + \overline{K}Q_n \quad (5.1)$$
- **Operation:** $J = 0, K = 0 \implies \text{Hold}$; $J = 0, K = 1 \implies \text{Reset (0)}$; $J = 1, K = 0 \implies \text{Set (1)}$; $J = 1, K = 1 \implies \text{Toggle } (\overline{Q}_n)$.

3. Clocked D Flip-Flop (Delay / Data)

- **Characteristic Equation:**

$$Q_{n+1} = D \quad (5.2)$$
- **Operation:** Directly transfers the input bit D to the output Q on the clock edge, introducing exactly one clock cycle of delay.

4. Clocked T Flip-Flop (Toggle)

Formed by tying the J and K inputs of a JK flip-flop together ($J = K = T$).

- **Characteristic Equation:**

$$Q_{n+1} = T \oplus Q_n = T\overline{Q}_n + \overline{T}Q_n \quad (5.3)$$
- **Operation:** When $T = 0$, output holds ($Q_{n+1} = Q_n$); when $T = 1$, output toggles ($Q_{n+1} = \overline{Q}_n$).

Present State $Q_n \rightarrow Q_{n+1}$	SR Inputs $S \quad R$		JK Inputs $J \quad K$		D Input D	T Input T
$0 \rightarrow 0$	0	X	0	X	0	0
$0 \rightarrow 1$	1	0	1	X	1	1
$1 \rightarrow 0$	0	1	X	1	0	1
$1 \rightarrow 1$	X	0	X	0	1	0

Table 5.3: Master Excitation Table for all four primary flip-flop types.

5.4 The Race-Around Condition & Master-Slave Architecture

5.4.1 The Race-Around Phenomenon

In a level-triggered JK flip-flop, when $J = 1$ and $K = 1$, the flip-flop operates in toggle mode. If the clock pulse width (t_p) is greater than the internal propagation delay of the flip-flop (t_{pd}):

$$t_{pd} < t_p < T \quad (5.4)$$

The output Q will toggle to $\overline{Q}$. Since the clock is still HIGH (1), this new output feeds back to the inputs, causing the flip-flop to toggle again and again continuously during the duration of t_p . At the end of the clock pulse, the final state of Q is indeterminate and unpredictable. This catastrophic timing glitch is called the **Race-Around Condition**.

5.4.2 The Master-Slave JK Flip-Flop Solution

The Master-Slave configuration completely eliminates race-around by decoupling input sampling from output state updates through two cascaded stages operated by complementary clock signals.

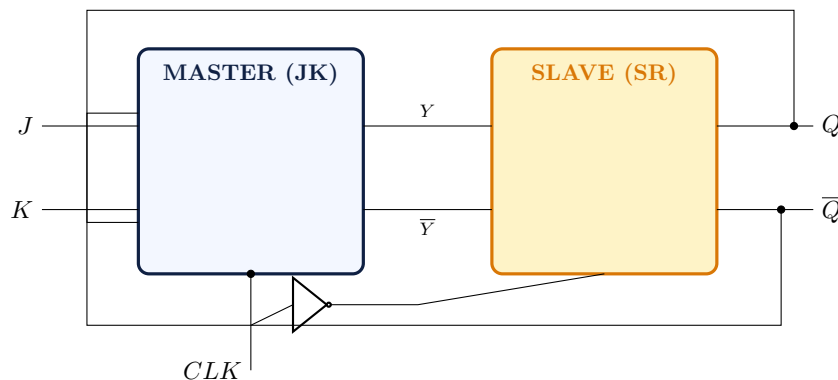


Figure 5.4: Master-Slave JK Flip-Flop architectural schematic.

- **Phase 1: Clock is HIGH ($CLK = 1$):** The Master is enabled and responds to inputs J and K , updating internal latch nodes Y and $\bar{Y}$. The Slave is disabled (since its inverted clock is 0) and holds its previous output state.
- **Phase 2: Clock Transitions to LOW ($CLK = 0$):** The Master is disabled and isolates its inputs. The Slave is enabled, reads the stored values from Y and $\bar{Y}$, and transfers them to the final outputs Q and $\bar{Q}$.
- **Result:** State transfer occurs cleanly on the falling edge of the clock pulse without feedback during the transition, permanently eliminating race-around.

5.5 Systematic Flip-Flop Conversion Methodology

5.5.1 Universal 5-Step Conversion Protocol

Any available source flip-flop can be converted to behave as any desired target flip-flop:

1. **Step 1: Identify Target and Available Flip-Flops.**
2. **Step 2: Construct the Conversion Table:** Tabulate Target Inputs, Present State (Q_n), and Next State (Q_{n+1}) from the Target Characteristic Table.
3. **Step 3: Determine Available Inputs:** Use the ****Excitation Table**** of the Available Flip-Flop to find required inputs for each transition $Q_n \rightarrow Q_{n+1}$.
4. **Step 4: K-Map Simplification:** Derive minimal expressions for the Available Flip-Flop inputs in terms of Target Inputs and Q_n .
5. **Step 5: Logic Circuit Synthesis:** Connect the derived combinational logic to the available flip-flop terminals.

Worked Example: Conversion: SR Flip-Flop to JK Flip-Flop

Convert an available SR Flip-Flop into a JK Flip-Flop.

Step 1: Conversion Table

Target (JK)		Present	Next	Available (SR)	
J	K	Q_n	Q_{n+1}	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1

Step 2: K-Map Minimization for S (J, K, Q_n):

- $S = 1$ at $(1, 0, 0)$ and $(1, 1, 0) \Rightarrow m_4, m_6$.
- $S = X$ at $(0, 0, 1)$ and $(1, 0, 1) \Rightarrow d_1, d_5$.
- Grouping m_4, m_6 yields: $\mathbf{S = J\bar{Q}_n}$.

Step 3: K-Map Minimization for R (J, K, Q_n):

- $R = 1$ at $(0, 1, 1)$ and $(1, 1, 1) \Rightarrow m_3, m_7$.
- $R = X$ at $(0, 0, 0)$ and $(0, 1, 0) \Rightarrow d_0, d_2$.
- Grouping m_3, m_7 yields: $\mathbf{R = KQ_n}$.

Final Implementation Equations:

$$\mathbf{S = J \cdot \bar{Q}_n, \quad R = K \cdot Q_n}$$

5.6 Summary & Chapter Review

Quick Revision: Unit 5 Sequential Logic Mastery Points

- **Latches vs. Flip-Flops:** Latches are level-triggered and transparent; Flip-flops are edge-triggered ($\uparrow$ or $\downarrow$).
- **SR Latch:** $S = R = 1$ forbidden in NOR latch; $\bar{S} = \bar{R} = 0$ forbidden in NAND latch.
- **Flip-Flop Characteristic Equations:**
 - SR: $Q_{n+1} = S + \bar{R}Q_n$ ($SR = 0$)
 - JK: $Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$
 - D: $Q_{n+1} = D$
 - T: $Q_{n+1} = T \oplus Q_n$
- **Race-Around Condition:** Occurs in level-triggered JK flip-flops when $J = K = 1$ and $t_p > t_{pd}$.
- **Master-Slave Solution:** Dual-stage architecture clocked on complementary phases

eliminates race-around.

- **Conversion Method:** Target Characteristic Table → Available Excitation Table → K-Map Optimization.

Applications of Sequential Circuits: Registers & Counters

6.1 Introduction

Registers and counters constitute the computational core of digital systems. While registers store and spatially/temporally transform binary words, counters generate deterministic numerical sequences, divide clock frequencies, and orchestrate microprocessor instruction cycles.

6.2 Shift Registers

An N -bit register consists of N cascaded flip-flops capable of storing an N -bit binary word. A **Shift Register** is capable of shifting binary data laterally (left or right) upon successive clock pulses.

6.2.1 The Four Primary Shift Register Topologies

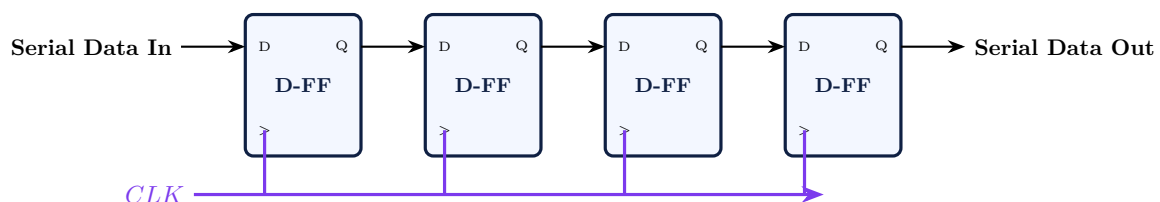


Figure 6.1: 4-bit Serial-In Serial-Out (SISO) Shift Register schematic using D Flip-Flops.

1. Serial-In Serial-Out (SISO)

- **Data Entry:** Single bit per clock cycle via a single input line.
- **Data Retrieval:** Single bit per clock cycle via the last flip-flop's output.
- **Timing Overhead:** For an N -bit word, it requires N clock pulses to fully load and an additional N clock pulses to shift out.

2. Serial-In Parallel-Out (SIPO)

- **Data Entry:** Serial bit stream into the first stage.
- **Data Retrieval:** All N bits are accessed simultaneously across parallel output pins ($Q_0, Q_1, \dots, Q_{N-1}$).
- **Application:** Serial-to-Parallel digital communications (e.g., UART receiver, SPI slave shift registers).

3. Parallel-In Serial-Out (PISO)

- **Data Entry:** Loaded simultaneously onto all flip-flops using combinational steering gates controlled by a **Shift/Load** signal.
- **Data Retrieval:** Shifted out serially one bit per clock cycle.
- **Application:** Parallel-to-Serial conversion (e.g., converting microprocessor data bus bytes into serial USB/UART transmissions).

4. Parallel-In Parallel-Out (PIPO)

- **Data Entry & Retrieval:** All bits are loaded in parallel and read out in parallel.
- **Timing Overhead:** Requires only **1 clock pulse** to store and update the entire N -bit register word.
- **Application:** General-purpose CPU data buffer registers and memory latch stages.

6.3 Binary Counters

A counter is a sequential state machine that transitions through a predefined sequence of binary states upon receiving clock pulses.

Definition: Counter Modulus (MOD-N)

The modulus of a counter is the total number of unique states it cycles through before repeating its sequence. An n -flip-flop counter has a maximum modulus of $\text{MOD-}2^n$.

6.3.1 Asynchronous (Ripple) Counters

In an Asynchronous Counter, the external clock signal is applied **only to the first (LSB) flip-flop**. Each subsequent flip-flop is clocked by the output transition of the preceding stage.

1. 3-Bit Asynchronous Ripple UP Counter

- **Construction:** 3 JK (or T) flip-flops with $J = K = 1$ (toggle mode).
- **Clocking Rule:** Negative-edge-triggered ($\downarrow$) flip-flops clocked from the **True Output (Q)** of the preceding stage count in **UP mode** ($0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 0$).

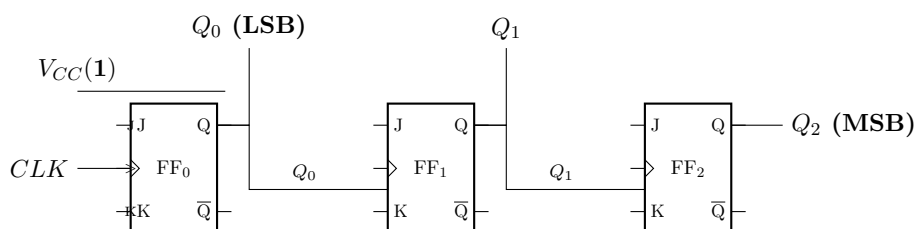


Figure 6.2: Logic circuit of a 3-Bit Asynchronous (Ripple) UP Counter.

2. Asynchronous DOWN Counter

- Clocking negative-edge-triggered flip-flops from the **Inverted Output ($\bar{Q}$)** of the preceding stage results in a **DOWN Counter** ($7 \rightarrow 6 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0 \rightarrow 7$).

3. Truncated Modulo- N Asynchronous Counters

To design a counter with modulus $N < 2^n$ (such as a **MOD-10 Decade / BCD Counter**):

1. Determine required flip-flops: $2^n \geq N \implies 2^4 \geq 10 \implies n = 4$.
2. Identify the binary state N where the counter must reset: $10_{10} = 1010_2$ ($Q_3 = 1, Q_2 = 0, Q_1 = 1, Q_0 = 0$).
3. Connect Q_3 and Q_1 to a **2-input NAND gate**, and feed its output to the active-low **CLR (Clear)** inputs of all four flip-flops.
4. When count reaches 1010_2 , the NAND output goes LOW, immediately forcing all flip-flops back to 0000_2 .

Common Pitfall: Cumulative Propagation Delay in Ripple Counters

In an N -bit ripple counter, each flip-flop introduces a propagation delay t_{pd} . The total time required for all bits to settle is $T_{total} = N \cdot t_{pd}$. To ensure correct counting without spurious output glitches:

$$T_{clock} \geq N \cdot t_{pd} \implies f_{max} \leq \frac{1}{N \cdot t_{pd}} \quad (6.1)$$

For high-frequency counting, **Synchronous Counters** must be used instead.

6.4 Synchronous Counters

In a **Synchronous Counter**, the master clock pulse is applied simultaneously to the clock inputs of all flip-flops, eliminating cumulative ripple delay.

6.4.1 3-Bit Synchronous UP Counter

- **Toggle Logic:** A stage toggles on clock pulse only if **all previous lower-order bits are 1**.

$$T_0 = 1 \quad (\text{FF}_0 \text{ toggles on every clock cycle})$$

$$T_1 = Q_0 \quad (\text{FF}_1 \text{ toggles when } Q_0 = 1)$$

$$T_2 = Q_0 \cdot Q_1 \quad (\text{FF}_2 \text{ toggles when both } Q_0 = 1 \text{ and } Q_1 = 1)$$

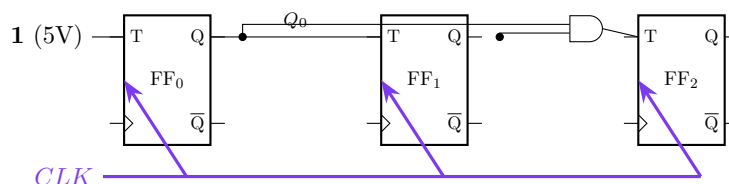


Figure 6.3: Complete circuit schematic of a 3-Bit Synchronous Parallel UP Counter.

Worked Example: Design of a Synchronous Mod-6 Counter using JK Flip-Flops

Design a Synchronous Counter that counts through the sequence: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 0$ using JK Flip-Flops.

Step 1: Determine Number of Flip-Flops Number of states $N = 6 \implies 2^n \geq 6 \implies n = 3$ Flip-Flops (Q_2, Q_1, Q_0). Unused states: 6(110₂) and 7(111₂).

Step 2: State Table and JK Excitation Mapping

Present State			Next State			FF ₂		FF ₁		FF ₀	
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	0	0	0	X	1	0	X	X	1

Step 3: K-Map Simplifications (with Don't Cares at 6 and 7):

- For FF₂: $J_2 = Q_1 Q_0$, $K_2 = Q_0$
- For FF₁: $J_1 = \overline{Q_2} Q_0$, $K_1 = Q_0$
- For FF₀: $J_0 = 1$, $K_0 = 1$

Step 4: Lock-Out / Self-Starting Verification

- **From Unused State 6(110₂):**

$$J_2 = 0, K_2 = 0 \implies Q_2^+ = 1$$

$$J_1 = 0, K_1 = 0 \implies Q_1^+ = 1$$

$$J_0 = 1, K_0 = 1 \implies Q_0^+ = 1 \implies \text{Next State is } \mathbf{7(111_2)}$$

- **From Unused State 7(111₂):**

$$J_2 = 1, K_2 = 1 \implies Q_2^+ = 0$$

$$J_1 = 0, K_1 = 1 \implies Q_1^+ = 0$$

$$J_0 = 1, K_0 = 1 \implies Q_0^+ = 0 \implies \text{Next State is } \mathbf{0(000_2)}$$

The counter automatically recovers to the valid counting cycle within 2 clock pulses $\implies$ ****Self-Starting Counter****.

6.5 Shift Register Counters (Ring & Johnson Counters)

6.5.1 Ring Counter

A Ring Counter is a circular shift register where the output of the final stage is fed directly to the input of the first stage ($Q_{\text{last}} \rightarrow D_{\text{first}}$).

- ****Modulus:**** MOD- N (N total states for N flip-flops).
- ****Initialization:**** Must be preset with a single **1** (e.g., 1000₂).
- ****Counting Cycle (4-bit):****

$$1000 \rightarrow 0100 \rightarrow 0010 \rightarrow 0001 \rightarrow 1000 \rightarrow \dots \quad (6.2)$$

- **Decoding:** Inherent (each flip-flop output directly indicates a specific sequence state without external decoding gates).

6.5.2 Johnson (Twisted Ring) Counter

A Johnson Counter connects the **inverted output** of the final stage to the input of the first stage ($\bar{Q}_{\text{last}} \rightarrow D_{\text{first}}$).

- **Modulus:** MOD- $2N$ ($2N$ total states for N flip-flops).
- **Initialization:** Typically cleared to all zeros (0000_2).
- **Counting Cycle (4-bit, 8 States):**

$$0000 \rightarrow 1000 \rightarrow 1100 \rightarrow 1110 \rightarrow 1111 \rightarrow 0111 \rightarrow 0011 \rightarrow 0001 \rightarrow 0000 \quad (6.3)$$

- **Decoding:** Decoded using simple 2-input AND gates ($S_0 = \bar{Q}_3\bar{Q}_0, S_1 = Q_0\bar{Q}_1, \dots$).

Counter Feature	Standard Counter	Binary Ring Counter	Johnson Counter
Number of States	2^N	N	$2N$
Clock Synchronization	Asynchronous or Synchronous	Synchronous	Synchronous
Decoding Logic	N -input AND gates	None (Direct outputs)	2-input AND gates
Unused States (4-bit)	0	12 unused states	8 unused states
Primary Applications	Arithmetic counting, timebases	Multiphase clocking, stepper motors	Frequency synthesis, timing generators

Table 6.1: Comprehensive Comparison of Counter Architectures.

6.6 Summary & Chapter Review

Quick Revision: Unit 6 Takeaways

- **Shift Register Topologies:** SISO (N in, N out), SIPO (serial to parallel), PISO (parallel to serial), PIPO (instantaneous 1-clock buffer).
- **Ripple Counters:** Asynchronous clocking; simple hardware but limited by cumulative propagation delay ($f_{\text{max}} \leq \frac{1}{N \cdot t_{pd}}$).
- **Synchronous Counters:** Clock connected to all flip-flops simultaneously; high operating speed.
- **Ring Counter:** N flip-flops $\Rightarrow N$ states; circular shift of a single bit **1**.
- **Johnson Counter:** N flip-flops $\Rightarrow 2N$ states; inverted feedback from $\bar{Q}_{\text{last}}$ to D_{first} .

Master Revision Sheet & Formula Reference

7.1 Introduction

This chapter provides a high-density, authoritative reference sheet designed for rapid revision before university examinations. It condenses all essential formulas, operational tables, conversion protocols, and circuit rules from Units 1 through 6.

7.2 Unit 1: DC Circuit Laws, Semiconductors, Diodes & BJTs

7.3 Unit 2: Arduino UNO, Signals & Sensor Interfacing

7.4 Unit 3: Number Systems, Binary Codes, Logic Gates & K-Maps

7.5 Unit 4: Combinational Logic Circuits

7.6 Unit 5 & 6: Sequential Logic, Flip-Flops, Registers & Counters

Law / Component	Formula / Relationship	Key Physical Context
Ohm's Law	$V = I \cdot R$	Constant temperature; linear ohmic materials only.
Resistivity Formula	$R = \rho \frac{L}{A} = \frac{L}{\sigma A}$	Geometric resistance derivation.
KCL (Node Rule)	$\sum I_{\text{in}} = \sum I_{\text{out}} \iff \sum I = 0$	Conservation of Electric Charge.
KVL (Loop Rule)	$\sum V_{\text{rise}} = \sum V_{\text{drop}} \iff \sum V = 0$	Conservation of Energy.
Voltage Division (Series)	$V_1 = V_s \left(\frac{R_1}{R_1 + R_2} \right)$	Applies strictly to series-connected resistors.
Current Division (Parallel)	$I_1 = I_s \left(\frac{R_2}{R_1 + R_2} \right)$	Applies strictly to two parallel branches.
Energy Band Gap (E_g)	Si: 1.12 eV, Ge: 0.72 eV	Insulator (> 5 eV), Conductor (0 eV).
Mass-Action Law	$n \cdot p = n_i^2$	Holds in equilibrium for intrinsic and extrinsic semiconductors.
Extrinsic Doping	N-Type: Pentavalent ($n \approx N_D$) P-Type: Trivalent ($p \approx N_A$)	Both types remain electrically neutral overall.
Shockley Diode Eq.	$I = I_0 \left(e^{\frac{V}{\eta V_T}} - 1 \right)$	$V_T = \frac{k_B T}{q} \approx 26 \text{ mV at } 300 \text{ K}.$
Diode Knee Voltage	Si: 0.7 V, Ge: 0.3 V	Cut-in forward conduction potential.
Half-Wave Rectifier	$V_{dc} = \frac{V_m}{\pi}$, $\gamma = 1.21$, $\eta = 40.6\%$, $\text{PIV} = V_m$	Single diode; conducts on positive half-cycle.
Bridge Rectifier	$V_{dc} = \frac{2V_m}{\pi}$, $\gamma = 0.482$, $\eta = 81.2\%$, $\text{PIV} = V_m$	4 diodes; no center-tap transformer required.
Center-Tap Rectifier	$V_{dc} = \frac{2V_m}{\pi}$, $\gamma = 0.482$, $\eta = 81.2\%$, $\text{PIV} = 2V_m$	2 diodes; center-tapped transformer.
BJT Current Law	$I_E = I_B + I_C$	$I_B \approx 1\text{--}5\% \implies I_E \approx I_C.$
BJT Current Gains	$\beta = \frac{I_C}{I_B}$, $\alpha = \frac{I_C}{I_E}$, $\beta = \frac{\alpha}{1-\alpha}$	$\beta \in [20, 500]$, $\alpha \in [0.95, 0.998].$
BJT Operating Modes	Cut-off: J_E Rev, J_C Rev (Switch OFF) Active: J_E Fwd, J_C Rev (Amplifier) Saturation: J_E Fwd, J_C Fwd (Switch ON)	$V_{CE(\text{sat})} \approx 0.2 \text{ V}$, $V_{BE} \approx 0.7 \text{ V}$ in active/sat.

Table 7.1: Comprehensive Formula Reference: DC Circuits, Semiconductors, Diodes, and BJTs.

Hardware Subsystem	Specification / Formula	Operational Context
ATmega328P Core	8-bit AVR RISC @ 16 MHz, 5V	32KB Flash, 2KB SRAM, 1KB EEPROM.
Power Inputs	USB (5V), DC Jack / VIN (7–12 V)	On-board 5V and 3.3V regulators.
10-bit SAR ADC (A0–A5)	$V_{in} = \text{ADC_Val} \times \left(\frac{5.0}{1023}\right)$	$2^{10} = 1024$ levels (0–1023), Step = 4.88 mV.
8-bit PWM (3, 5, 6, 9, 10, 11)	$V_{avg} = 5.0 \times \left(\frac{\text{PWM_Val}}{255}\right)$	Pins 5, 6: ≈ 980 Hz; Pins 3, 9, 10, 11: ≈ 490 Hz.
Active IR Sensor	IR LED + Photodiode + LM393	White reflects (Out LOW); Black absorbs (Out HIGH).
LDR Photoresistor	$V_{out} = V_{CC} \left(\frac{R_{fixed}}{R_{LDR} + R_{fixed}} \right)$	CdS photoconductivity; light decreases resistance.
Ultrasonic (HC-SR04)	$D = \frac{v \cdot \Delta t}{2} = \frac{\Delta t(\mu s)}{58.3} \text{ cm}$	$10 \mu s$ Trig $\rightarrow$ 8-cycle 40 kHz burst $\rightarrow$ Echo pulse.
DHT11 vs. DHT22	DHT11: 20–90% RH, 0–50°C, 1 Hz DHT22: 0–100% RH, –40–80°C, 0.5 Hz	40-bit single-bus serial digital packet.

Table 7.2: Arduino UNO Subsystems and Sensor Interfacing Reference Table.

Topic Domain	Algorithm / Mathematical Rule	Example / Significance
Decimal $\rightarrow$ Base r	Successive division (int), successive mult (frac).	$(25.625)_{10} = (11001.101)_2$.
Binary $\leftrightarrow$ Octal / Hex	Group into 3 bits (Octal) or 4 bits (Hex).	$(110110)_2 = (66)_8 = (36)_{16}$.
BCD (8421 Code)	4-bit representation of digits 0–9.	1010_2 – 1111_2 are invalid; add 0110_2 .
Excess-3 Code (XS-3)	$\text{BCD} + 0011_2$.	Self-complementing code.
Binary $\rightarrow$ Gray Code	$G_{n-1} = B_{n-1}, \quad G_i = B_{i+1} \oplus B_i$	Unit-distance code; eliminates glitching.
Gray $\rightarrow$ Binary Code	$B_{n-1} = G_{n-1}, \quad B_i = B_{i+1} \oplus G_i$	Sequential XOR decoding.
2's Comp. Subtraction	$A - B = A + (\overline{B} + 1)$	Carry = 1 $\implies$ (+); Carry = 0 $\implies$ (–).
De Morgan's Laws	$\overline{A \cdot B} = \overline{A} + \overline{B}, \quad \overline{A + B} = \overline{A} \cdot \overline{B}$	Fundamental duality theorem.
Absorption Laws	$A + AB = A, \quad A(A + B) = A, \quad A + \overline{A}B = A + B$	Key Boolean minimization tool.
Canonical Forms	SOP: $F = \sum m(\dots), \quad \text{POS: } F = \prod M(\dots)$	$m_i = \overline{M}_i$.
K-Map Grouping	Group sizes 1, 2, 4, 8, 16. Gray code layout.	Corner Quad $(m_0, m_2, m_8, m_{10}) = \overline{B} \overline{D}$.

Table 7.3: Number Systems, Codes, and Boolean Minimization Reference Table.

Circuit Block	Output Boolean Expressions	Structural Implementation
Half Adder	$S = A \oplus B, \quad C = A \cdot B$	1 XOR gate, 1 AND gate.
Full Adder	$S = A \oplus B \oplus C_{in}, \quad C_{out} = AB + C_{in}(A \oplus B)$	2 Half Adders + 1 OR gate.
Half Subtractor	$D = A \oplus B, \quad B_{out} = \overline{A} \cdot B$	1 XOR gate, 1 AND gate, 1 NOT gate.
Full Subtractor	$D = A \oplus B \oplus B_{in}, \quad B_{out} = \overline{A}B + B_{in}(\overline{A} \oplus \overline{B})$	2 Half Subtractors + 1 OR gate.
4:1 Multiplexer	$Y = \overline{S_1}\overline{S_0}I_0 + \overline{S_1}S_0I_1 + S_1\overline{S_0}I_2 + S_1S_0I_3$	4 AND gates, 1 OR gate, 2 NOT gates.
1:4 De-multiplexer	$Y_k = S_{term} \cdot D_{in}$	4 AND gates, 2 NOT gates.
2-to-4 Decoder	$Y_0 = E\overline{A}\overline{B}, \quad Y_1 = E\overline{A}B, \quad Y_2 = EA\overline{B}, \quad Y_3 = EAB$	Active-High or Active-Low (74138).
8-to-3 Octal Encoder	$X = D_4 + D_5 + D_6 + D_7, \quad Y = D_2 + D_3 + D_6 + D_7, \quad Z = D_1 + D_3 + D_5 + D_7$	3 OR gates. Priority encoders resolve collisions.
2-Bit Comparator	$x_1 = \overline{A_1} \oplus \overline{B_1}, \quad x_0 = \overline{A_0} \oplus \overline{B_0}$ $(A = B) = x_1x_0$ $(A > B) = \overline{A_1}\overline{B_1} + x_1\overline{A_0}\overline{B_0}$ $(A < B) = \overline{A_1}B_1 + x_1\overline{A_0}B_0$	Standard XNOR, AND, OR network.

Table 7.4: Master Summary Table of Combinational Logic Blocks.

Flip-Flop Type	Characteristic Equation	Excitation Table ($Q_n \rightarrow Q_{n+1}$)
SR Flip-Flop	$Q_{n+1} = S + \overline{R}Q_n \quad (SR = 0)$	$0 \rightarrow 0 : 0X, 0 \rightarrow 1 : 10, 1 \rightarrow 0 : 01, 1 \rightarrow 1 : X0$
JK Flip-Flop	$Q_{n+1} = J\overline{Q}_n + \overline{K}Q_n$	$0 \rightarrow 0 : 0X, 0 \rightarrow 1 : 1X, 1 \rightarrow 0 : X1, 1 \rightarrow 1 : X0$
D Flip-Flop	$Q_{n+1} = D$	$0 \rightarrow 0 : 0, 0 \rightarrow 1 : 1, 1 \rightarrow 0 : 0, 1 \rightarrow 1 : 1$
T Flip-Flop	$Q_{n+1} = T \oplus Q_n$	$0 \rightarrow 0 : 0, 0 \rightarrow 1 : 1, 1 \rightarrow 0 : 1, 1 \rightarrow 1 : 0$

Table 7.5: Characteristic and Excitation Table Summary for all Flip-Flops.

Sequential Subsystem	Operational Principle	Performance / Modulus
Master-Slave JK FF	Dual-stage clocked on complementary clock levels.	Eliminates Race-Around glitch ($t_p > t_{pd}$).
SISO Shift Register	Serial in, serial out via cascaded D-FFs.	N cycles to load, N cycles to shift out.
SIPO Shift Register	Serial in, parallel readout on bus.	N cycles to load, 0 cycles to read.
PISO Shift Register	Parallel load via Shift/Load gate, serial out.	Parallel-to-serial conversion.
PIPO Shift Register	Parallel load, parallel read.	1 clock pulse to store and read.
Asynchronous (Ripple) Counter	Clock ripples from $Q_k \rightarrow CLK_{k+1}$.	Low speed ($f_{max} \leq \frac{1}{N \cdot t_{pd}}$); Mod- N via NAND clear.
Synchronous Counter	Common clock to all FFs; look-ahead AND gates.	High speed ($f_{max} = \frac{1}{t_{pd} + t_{gate}}$); glitch-free.
Ring Counter	Circular shift ($Q_{last} \rightarrow D_{first}$); initialized with 1000.	MOD- N (N states for N flip-flops).
Johnson Counter	Inverted feedback ($\overline{Q}_{last} \rightarrow D_{first}$); initialized 0000.	MOD- $2N$ ($2N$ states for N flip-flops).

Table 7.6: Summary Comparison of Sequential Registers and Counters.

Comprehensive Solved Practice Problems

8.1 Introduction

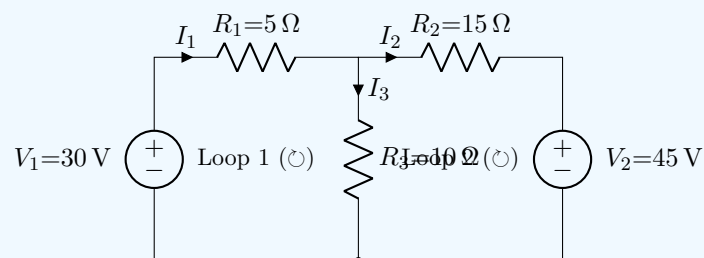
This chapter provides extensive, multi-tiered solved practice problems covering all six units of the ECE249 curriculum. Each problem follows the standardized engineering solution protocol:

Given $\rightarrow$ Required $\rightarrow$ Principle/Formula $\rightarrow$ Step-by-Step Calculation $\rightarrow$ Final Answer

8.2 Unit 1: Electrical Laws, Semiconductors, Diodes & BJTs

Worked Example: Multi-Loop DC Circuit Analysis via Mesh Current Method

Problem Statement: In the dual-source DC circuit shown below, determine the branch currents I_1, I_2, I_3 and calculate the total power dissipated in the 10Ω central load resistor.



Step 1: Given Data

- $V_1 = 30\text{ V}, \quad V_2 = 45\text{ V}$
- $R_1 = 5\Omega, \quad R_2 = 15\Omega, \quad R_3 = 10\Omega$

Step 2: Nodal KCL Equation At top node: $I_1 = I_2 + I_3 \Rightarrow I_3 = I_1 - I_2$.

Step 3: Loop 1 KVL (Clockwise)

$$\begin{aligned}
 +V_1 - I_1 R_1 - I_3 R_3 &= 0 \\
 30 - 5I_1 - 10(I_1 - I_2) &= 0 \\
 30 - 15I_1 + 10I_2 &= 0 \Rightarrow 15I_1 - 10I_2 = 30 \Rightarrow 3I_1 - 2I_2 = 6 \quad \text{--- (1)}
 \end{aligned}$$

Step 4: Loop 2 KVL (Clockwise)

$$\begin{aligned}
 +I_3R_3 - I_2R_2 - V_2 &= 0 \\
 10(I_1 - I_2) - 15I_2 - 45 &= 0 \\
 10I_1 - 25I_2 &= 45 \implies 2I_1 - 5I_2 = 9 \quad \text{--- (2)}
 \end{aligned}$$

Step 5: Solve System of Equations Multiply (1) by 5 and (2) by 2:

$$\begin{aligned}
 15I_1 - 10I_2 &= 30 \\
 4I_1 - 10I_2 &= 18 \\
 \text{Subtract: } 11I_1 &= 12 \implies I_1 = \frac{12}{11} \approx 1.091 \text{ A}
 \end{aligned}$$

Substitute I_1 into (1):

$$3\left(\frac{12}{11}\right) - 2I_2 = 6 \implies \frac{36}{11} - 6 = 2I_2 \implies 2I_2 = -\frac{30}{11} \implies I_2 = -\frac{15}{11} \approx -1.364 \text{ A}$$

Current through R_3 :

$$I_3 = I_1 - I_2 = \frac{12}{11} - \left(-\frac{15}{11}\right) = \frac{27}{11} \approx 2.455 \text{ A}$$

Step 6: Power Dissipated in R_3

$$P_{R_3} = I_3^2 \cdot R_3 = (2.4545)^2 \times 10 = 60.25 \text{ W}$$

Final Answers:

$$\mathbf{I_1 = 1.09 \text{ A}, \quad I_2 = -1.36 \text{ A}, \quad I_3 = 2.45 \text{ A}, \quad P_{R_3} = 60.25 \text{ W}}$$

Note: The negative sign for I_2 indicates physical current flows counter-clockwise, opposite to the assumed mesh direction.

Worked Example: Full-Wave Bridge Rectifier with Silicon Diodes

Problem Statement: A Full-Wave Bridge Rectifier is connected to a $230 \text{ V}_{\text{rms}}$, 50 Hz AC supply through a step-down transformer of turns ratio $10 : 1$. The load resistance is $R_L = 100 \Omega$. Each silicon diode has a forward voltage drop $V_D = 0.7 \text{ V}$ and dynamic resistance $r_f = 1.5 \Omega$.

1. Find the peak secondary voltage V_m .
2. Determine the DC output voltage V_{dc} and DC load current I_{dc} .
3. Calculate the ripple factor γ and rectification efficiency η .

Step 1: Given Data

- $V_{pri,rms} = 230 \text{ V}$, Turns ratio $N_1/N_2 = 10 \implies V_{sec,rms} = \frac{230}{10} = 23 \text{ V}$
- $R_L = 100 \Omega$, $V_D = 0.7 \text{ V}$, $r_f = 1.5 \Omega$

Step 2: Peak Voltage Calculation

$$V_m = V_{sec,rms} \times \sqrt{2} = 23 \times \sqrt{2} \approx 32.53 \text{ V}$$

Effective peak voltage across load taking two conducting diodes into account:

$$V_{m,\text{eff}} = V_m - 2V_D = 32.53 - 2(0.7) = 31.13 \text{ V}$$

Peak load current:

$$I_m = \frac{V_{m,\text{eff}}}{2r_f + R_L} = \frac{31.13}{2(1.5) + 100} = \frac{31.13}{103} \approx 0.3022 \text{ A} = 302.2 \text{ mA}$$

Step 3: DC Output Quantities

$$I_{dc} = \frac{2I_m}{\pi} = \frac{2 \times 0.3022}{\pi} \approx 0.1924 \text{ A} = 192.4 \text{ mA}$$

$$V_{dc} = I_{dc} \cdot R_L = 0.1924 \times 100 = 19.24 \text{ V}$$

Step 4: RMS Quantities, Ripple Factor and Efficiency

$$I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{0.3022}{\sqrt{2}} \approx 0.2137 \text{ A}$$

$$\gamma = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} = \sqrt{\left(\frac{0.2137}{0.1924}\right)^2 - 1} = \sqrt{(1.1107)^2 - 1} \approx 0.483$$

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 R_L}{I_{rms}^2 (2r_f + R_L)} = \frac{(0.1924)^2 \times 100}{(0.2137)^2 \times 103} = \frac{3.7018}{4.7037} \approx 78.7\%$$

Final Answers:

$$V_m = 32.53 \text{ V}, \quad V_{dc} = 19.24 \text{ V}, \quad I_{dc} = 192.4 \text{ mA}, \quad \gamma = \mathbf{0.483}, \quad \eta = \mathbf{78.7\%}$$

8.3 Unit 2: Arduino ADC, PWM & Sensors

Worked Example: Ultrasonic Rangefinder Distance with Temperature Correction

Problem Statement: An HC-SR04 ultrasonic sensor interfaced with an Arduino UNO detects an obstacle with an echo pulse duration $\Delta t = 1750 \mu\text{s}$. Ambient temperature is measured by a DHT11 sensor as $T = 30^\circ\text{C}$.

1. Compute the temperature-corrected velocity of sound in air using $v = 331.3 + 0.606 \cdot T (\text{m} \cdot \text{s}^{-1})$.
2. Determine the exact physical distance to the obstacle in centimeters.

Step 1: Given Data

- $\Delta t = 1750 \mu\text{s} = 1750 \times 10^{-6} \text{ s}$
- Temperature $T = 30^\circ\text{C}$

Step 2: Calculate Speed of Sound at $T = 30^\circ\text{C}$

$$v_{\text{sound}} = 331.3 + 0.606(30) = 331.3 + 18.18 = 349.48 \text{ m} \cdot \text{s}^{-1} = 0.03495 \text{ cm} \cdot \mu\text{s}^{-1}$$

Step 3: Calculate Target Distance

$$D = \frac{v_{\text{sound}} \times \Delta t}{2} = \frac{0.034948 \text{ cm}/\mu\text{s} \times 1750 \mu\text{s}}{2} = \frac{61.159}{2} \approx 30.58 \text{ cm}$$

Final Answers:

$$v_{\text{sound}} = 349.48 \text{ m} \cdot \text{s}^{-1}, \quad \text{Distance } \mathbf{D} = 30.58 \text{ cm}$$

8.4 Unit 3: Number Systems, Boolean Simplification & K-Maps

Worked Example: Multi-Base Number Conversions**Problem Statement:** Convert the binary number $N = (11010110.1011)_2$ into:

1. Octal
2. Hexadecimal
3. Decimal

Part 1: Binary to Octal (Group in 3s)

- Integer: $\underbrace{011}_3 \underbrace{010}_2 \underbrace{110}_6 \Rightarrow (326)_8$
- Fraction: $\underbrace{101}_5 \underbrace{100}_4 \Rightarrow (.54)_8$
- $N = (326.54)_8$

Part 2: Binary to Hexadecimal (Group in 4s)

- Integer: $\underbrace{1101}_D \underbrace{0110}_6 \Rightarrow (D6)_{16}$
- Fraction: $\underbrace{1011}_B \Rightarrow (.B)_{16}$
- $N = (D6.B)_{16}$

Part 3: Hexadecimal to Decimal

$$\begin{aligned} N &= 13 \times 16^1 + 6 \times 16^0 + 11 \times 16^{-1} \\ &= 208 + 6 + \frac{11}{16} = 214 + 0.6875 = (214.6875)_{10} \end{aligned}$$

Final Answer: $(214.6875)_{10}$ **Worked Example: K-Map Minimization with Don't Cares****Problem Statement:** Minimize the following Boolean expression and realize it using only NAND gates:

$$F(A, B, C, D) = \sum m(0, 1, 4, 8, 9, 12) + d(2, 10)$$

Step 1: Plotting K-Map ($AB \setminus CD$)

- 1s at: $m_0(0000), m_1(0001), m_4(0100), m_8(1000), m_9(1001), m_{12}(1100)$
- Don't Cares at: $d_2(0010), d_{10}(1010)$

Step 2: Grouping Terms

1. **Octet / Quad in Columns 00 and 01:** Notice cells (m_0, m_1, m_8, m_9) form a Quad $\Rightarrow \overline{B}\overline{C}$.
2. **Quad in Column 00 (m_0, m_4, m_{12}, m_8):** All 4 rows in column $CD = 00 \Rightarrow \overline{C}\overline{D}$.
3. **Check cells (m_0, d_2, m_8, d_{10}):** Form a Quad in Rows 00, 10 and Cols 00, 10 $\Rightarrow \overline{B}\overline{D}$.

Minimal SOP expression:

$$F = \overline{B}\overline{C} + \overline{C}\overline{D}$$

(Or factoring: $F = \overline{C}(\overline{B} + \overline{D}) = \overline{C}\overline{B}\overline{D}$).

Step 3: NAND-Only Implementation Applying double complement:

$$F = \overline{\overline{\overline{B}\overline{C} + \overline{C}\overline{D}}} = \overline{(\overline{B}\overline{C}) \cdot (\overline{C}\overline{D})}$$

Synthesized using 4 NAND gates.

8.5 Unit 4: Combinational Logic Circuits

Worked Example: Boolean Function Synthesis with 4:1 Multiplexer

Problem Statement: Implement the logic function $F(A, B, C) = \sum m(1, 2, 6, 7)$ using a single 4:1 Multiplexer and no external logic gates. Use A and B as the select lines ($S_1 = A, S_0 = B$).

Step 1: Implementation Table With $S_1 = A, S_0 = B$, the remaining variable C provides the data inputs:

Input Line	$S_1S_0 = 00$	$S_1S_0 = 01$	$S_1S_0 = 10$	$S_1S_0 = 11$	Data Input Assignment
$\overline{C}$	$m_0(0)$	$m_2(1)$	$m_4(0)$	$m_6(1)$	$I_0 = C$ $I_1 = \overline{C}$ $I_2 = 0$ $I_3 = 1$
C	$m_1(1)$	$m_3(0)$	$m_5(0)$	$m_7(1)$	

Step 2: Input Evaluations

- For $S_1S_0 = 00$: m_1 is circled $\Rightarrow I_0 = C$.
- For $S_1S_0 = 01$: m_2 is circled $\Rightarrow I_1 = \overline{C}$.
- For $S_1S_0 = 10$: Neither is circled $\Rightarrow I_2 = 0$ (GND).
- For $S_1S_0 = 11$: Both m_6, m_7 are circled $\Rightarrow I_3 = 1$ (VCC).

Final Connections:

$$S_1 = A, \quad S_0 = B, \quad I_0 = C, \quad I_1 = \overline{C}, \quad I_2 = 0, \quad I_3 = 1$$

8.6 Unit 5: Sequential Flip-Flop Conversions

Worked Example: Flip-Flop Conversion: JK to D Flip-Flop

Problem Statement: Convert an available JK Flip-Flop into a D Flip-Flop.

Step 1: Conversion Table

Target Input (D)	Present State (Q_n)	Next State (Q_{n+1})	Available Inputs (J, K)	
	Q_n	Q_{n+1}	J	K
0	0	0	0	X
0	1	0	X	1
1	0	1	1	X
1	1	1	X	0

Step 2: K-Map Simplification

- For $J(D, Q_n)$: $J = 1$ at $(1, 0)$, $J = X$ at $(0, 1)$ and $(1, 1) \Rightarrow \mathbf{J = D}$.
- For $K(D, Q_n)$: $K = 1$ at $(0, 1)$, $K = X$ at $(0, 0)$ and $(1, 0) \Rightarrow \mathbf{K = \overline{D}}$.

Step 3: Implementation Schematic Connect input D directly to terminal J . Connect an inverter between D and terminal K ($K = \overline{D}$).

8.7 Unit 6: Synchronous Counter Design

Worked Example: Design of a Synchronous Mod-3 Counter using T Flip-Flops

Problem Statement: Design a synchronous counter that counts in the sequence $00_2 \rightarrow 01_2 \rightarrow 10_2 \rightarrow 00_2$ using T Flip-Flops.

Step 1: State Table and T-Excitation Mapping

Unused state: 11_2 .

Present State		Next State		T Inputs ($T = Q_n \oplus Q_{n+1}$)	
Q_1	Q_0	Q_1^+	Q_0^+	T_1	T_0
0	0	0	1	0	1
0	1	1	0	1	1
1	0	0	0	1	0
1	1	X	X	X	X

Step 2: K-Map Minimization (with Don't Care at $Q_1Q_0 = 11$)

- For T_1 : 1s at (0, 1) and (1, 0), X at (1, 1). Group (0, 1) with (1, 1) $\Rightarrow Q_0$; Group (1, 0) with (1, 1) $\Rightarrow Q_1$. $\mathbf{T_1 = Q_0 + Q_1}$.
- For T_0 : 1s at (0, 0) and (0, 1), X at (1, 1). Group (0, 0) with (0, 1) $\Rightarrow \overline{Q_1}$. $\mathbf{T_0 = \overline{Q_1}}$.

Step 3: Lock-out Verification From unused state 11_2 : $T_1 = 1 + 1 = 1 \Rightarrow Q_1^+ = 0$; $T_0 = \overline{1} = 0 \Rightarrow Q_0^+ = 1$. Next state is 01_2 (valid cycle) $\Rightarrow$ ****Self-Starting****.

University-Style Mock Examinations

9.1 Examination Strategy & Time Management

University end-term examinations for ECE249 are typically structured for ****100 Marks**** over a ****3-Hour (180 Minutes)**** duration.

Exam Tip: Optimal 180-Minute Exam Time Allocation

- **First 5 Minutes:** Question paper scan and selection of optimal optional questions.
- **Section A (20 Marks):** 10 Short questions $\times$ 2.5 min = **25 Minutes**.
- **Section B (30 Marks):** 6 Analytical problems $\times$ 9 min = **54 Minutes**.
- **Section C (50 Marks):** 5 Comprehensive long questions $\times$ 18 min = **90 Minutes**.
- **Final 10 Minutes:** Numerical verification, unit checking, circuit label inspection.

9.2 Mock Examination Paper 1

Course Code: ECE249

Course Title: Digital Electronics & Microcontroller Interfacing

Maximum Marks: 100

Time Allowed: 3 Hours

SECTION A: Short Conceptual Questions ($10 \times 2 = 20$ Marks)

Answer all questions. Each question carries 2 marks.

- Q1.** State Ohm's Law and name two non-ohmic devices where it fails.
- Q2.** State Kirchhoff's Current Law (KCL) and identify the fundamental physical conservation law upon which it is based.
- Q3.** Distinguish between Intrinsic and Extrinsic semiconductors with respect to charge carrier concentration.
- Q4.** Define the knee voltage of a PN junction diode. State its typical values for Silicon and Germanium.
- Q5.** Why does the Arduino UNO operate with an on-board crystal oscillator frequency of 16 MHz?
- Q6.** Convert the binary number $(101101)_2$ into its corresponding Gray code.
- Q7.** State De Morgan's first and second theorems algebraically.
- Q8.** Why are NAND and NOR gates designated as "Universal Gates"?

- Q9.** Define the Race-Around Condition in a level-triggered JK flip-flop.
- Q10.** Differentiate between a Ring Counter and a Johnson Counter in terms of their modulus for an N -flip-flop register.

SECTION B: Analytical & Numerical Problems ($6 \times 5 = 30$ Marks)

Answer any six questions. Each question carries 5 marks.

- Q11.** Use the Voltage Division and Current Division rules to find the voltage V_x and current I_x in the circuit containing a 48 V source with series resistors 12Ω and a parallel combination of 20Ω and 30Ω .
- Q12.** A full-wave bridge rectifier with ideal diodes delivers power to a load resistance $R_L = 1\text{ k}\Omega$ from an AC source $v_s(t) = 120\sqrt{2}\sin(120\pi t)\text{ V}$. Calculate: (a) V_{dc} , (b) I_{dc} , (c) Ripple voltage V_{ac} , and (d) Rectifier efficiency η .
- Q13.** An Arduino UNO ADC reads an analog voltage from an LDR connected in a pull-down voltage divider with a $10\text{ k}\Omega$ fixed resistor. If the ADC registers a digital value of 750 with $V_{ref} = 5.0\text{ V}$, calculate the measured voltage and the instantaneous resistance of the LDR.
- Q14.** Perform the subtraction $(42)_{10} - (68)_{10}$ using 8-bit 2's complement arithmetic. Clearly state the carry status and final signed magnitude.
- Q15.** Minimize the 4-variable logic function using a K-Map:

$$F(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + d(0, 2, 5)$$

- Q16.** Synthesize a Half Subtractor using only two-input NAND gates.
- Q17.** Convert an available D Flip-Flop into a T Flip-Flop. Show the complete conversion table, K-Map, and circuit diagram.

SECTION C: Comprehensive Design Questions ($5 \times 10 = 50$ Marks)

Answer all five questions. Each question carries 10 marks.

- Q18.** (a) Explain the construction and operation of an NPN Bipolar Junction Transistor in Common Emitter (CE) configuration.
(b) Sketch and explain the input and output V-I characteristics, clearly demarcating the Cut-off, Active, and Saturation regions.
(c) Derive the mathematical relationship between α and β .
- Q19.** (a) Draw the architectural block diagram of an Ultrasonic Distance Sensor (HC-SR04) and explain its operational timing protocol.
(b) Derive the exact distance formula and explain why division by 2 is required.
(c) If the echo pulse width measured by an Arduino microcontroller is $2500\text{ }\mu\text{s}$, calculate the physical obstacle distance in centimeters (assuming $v = 340\text{ m} \cdot \text{s}^{-1}$).
- Q20.** (a) Design a 4:1 Multiplexer using basic logic gates and write its complete Boolean equation.
(b) Implement the 4-variable Boolean function $F(A, B, C, D) = \sum m(0, 1, 3, 4, 8, 9, 15)$ using an 8:1 Multiplexer with A, B, C as select lines.
- Q21.** (a) What is the Race-Around condition in JK flip-flops? Derive the mathematical inequality that causes it.
(b) Explain with a complete schematic and timing diagrams how a Master-Slave JK Flip-Flop permanently eliminates the race-around condition.
- Q22.** Design a 3-bit Synchronous UP Counter using JK Flip-Flops.
(a) Draw the State Diagram and State Transition Table.
(b) Map the flip-flop excitations using the JK excitation table.
(c) Simplify all J and K input equations using K-Maps.
(d) Draw the final logic circuit schematic.

9.3 Solutions for Mock Examination Paper 1

Solutions for Section A

Solutions to Questions 1–10

- Ans Q1. Ohm's Law:** At constant temperature, current through a conductor is proportional to applied voltage ($V = IR$). Two non-ohmic devices: ****PN Junction Diode**** and ****Bipolar Junction Transistor (BJT)****.
- Ans Q2. KCL:** $\sum I_{\text{entering}} = \sum I_{\text{leaving}}$ at any node. It is based on the ****Conservation of Electric Charge****.
- Ans Q3. Semiconductors:** In Intrinsic semiconductors, $n = p = n_i$. In Extrinsic N-type, electrons dominate ($n \approx N_D \gg p$); in P-type, holes dominate ($p \approx N_A \gg n$).
- Ans Q4. Knee Voltage (V_k):** The forward bias voltage at which diode current begins to increase exponentially. $V_k(\text{Si}) = 0.7 \text{ V}$, $V_k(\text{Ge}) = 0.3 \text{ V}$.
- Ans Q5. 16 MHz Clock:** Established by a quartz crystal oscillator to provide precise instruction cycle execution timing (62.5 ns per clock cycle) for the ATmega328P processor.
- Ans Q6. Binary to Gray:** Binary $101101_2 \implies G_5 = 1, G_4 = 1 \oplus 0 = 1, G_3 = 0 \oplus 1 = 1, G_2 = 1 \oplus 1 = 0, G_1 = 1 \oplus 0 = 1, G_0 = 0 \oplus 1 = 1 \implies \mathbf{111011}_{\text{Gray}}$.
- Ans Q7. De Morgan's Theorems:** (1) $\overline{A \cdot B} = \overline{A} + \overline{B}$, (2) $\overline{A + B} = \overline{A} \cdot \overline{B}$.
- Ans Q8. Universal Gates:** Because any Boolean logic function (AND, OR, NOT, XOR, XNOR) can be implemented using exclusively NAND gates or exclusively NOR gates without other gate types.
- Ans Q9. Race-Around Condition:** In a level-triggered JK flip-flop, when $J = K = 1$ and clock pulse width $t_p > t_{pd}$ (propagation delay), the output toggles repeatedly and unpredictably during a single clock pulse.
- Ans Q10. Counter Modulus:** An N -flip-flop ****Ring Counter**** has modulus MOD- N (N states), whereas an N -flip-flop ****Johnson Counter**** has modulus MOD- $2N$ ($2N$ states).

Solutions for Section B

Solution to Q11: DC Circuit Division

Parallel equivalent of 20Ω and 30Ω :

$$R_p = \frac{20 \times 30}{20 + 30} = \frac{600}{50} = 12 \Omega$$

Total circuit resistance $R_{\text{total}} = R_1 + R_p = 12 + 12 = 24 \Omega$. Total source current $I_s =$

$\frac{V_s}{R_{total}} = \frac{48}{24} = 2.0 \text{ A}$. By Voltage Division Rule across parallel branch:

$$V_x = V_s \left(\frac{R_p}{R_1 + R_p} \right) = 48 \left(\frac{12}{12 + 12} \right) = 24 \text{ V}$$

By Current Division Rule through the 20Ω resistor:

$$I_x = I_s \left(\frac{30}{20 + 30} \right) = 2.0 \times \frac{30}{50} = 1.2 \text{ A}$$

Solution to Q12: Bridge Rectifier Analysis

Peak secondary voltage $V_m = 120\sqrt{2} \approx 169.7 \text{ V}$.

- (a) $V_{dc} = \frac{2V_m}{\pi} = \frac{2 \times 169.7}{\pi} \approx 108.04 \text{ V}$
- (b) $I_{dc} = \frac{V_{dc}}{R_L} = \frac{108.04}{1000 \Omega} = 0.108 \text{ A} = 108.04 \text{ mA}$
- (c) $V_{rms} = \frac{V_m}{\sqrt{2}} = 120 \text{ V} \Rightarrow V_{ac} = \sqrt{V_{rms}^2 - V_{dc}^2} = \sqrt{120^2 - 108.04^2} = 52.22 \text{ V}$
- (d) $\eta = \frac{8}{\pi^2} \approx 81.2\%$

Solution to Q14: 2's Complement Subtraction

Compute $(42)_{10} - (68)_{10}$ in 8 bits:

- $(42)_{10} = 00101010_2$, $(68)_{10} = 01000100_2$
- 1's complement of 68 = $10111011_2 \Rightarrow$ 2's complement of 68 = 10111100_2
- Add: $00101010_2 + 10111100_2 = 11100110_2$
- No end-around carry ($C_{out} = 0$) $\Rightarrow$ Result is Negative.
- Magnitude = 2's comp of $11100110_2 = 00011001_2 + 1 = 00011010_2 = (26)_{10}$.
- Result = $-(26)_{10}$ (11100110_2).

Solution to Q15: K-Map Minimization

$$F(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + d(0, 2, 5).$$

- Column $CD = 11$ contains $m_3, m_7, m_{15}, m_{11} \Rightarrow$ forms an ****Octet / Quad**** CD .
- Row $AB = 00$ contains $d_0, m_1, m_3, d_2 \Rightarrow$ forms a ****Quad**** $\overline{A}\overline{B}$.
- Remaining m_5 (with m_1, m_3, m_7) forms Quad $\overline{A}D$.
- Minimal equation: $\mathbf{F} = \overline{A}\overline{B} + \mathbf{CD}$ (or $\overline{A}D + CD$).

Solution to Q17: D to T Flip-Flop Conversion

Target is T-FF ($T, Q_n \rightarrow Q_{n+1}$); Available is D-FF ($Q_{n+1} = D$).

T	Q_n	Q_{n+1}	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

From the table, $D = 1$ when $(T, Q_n) = (0, 1)$ or $(1, 0) \implies \mathbf{D = T \oplus Q_n}$. Realization: Connect an XOR gate with inputs T and Q_n to the D input terminal.

9.4 Mock Examination Paper 2

Course Code: ECE249

Course Title: Digital Electronics & Microcontroller Interfacing

Maximum Marks: 100

Time Allowed: 3 Hours

SECTION A: Short Conceptual Questions ($10 \times 2 = 20$ Marks)

- Q1. Why does a silicon PN diode require 0.7 V to turn ON, whereas Germanium requires only 0.3 V?
- Q2. State the relationship between collector current I_C , base current I_B , and current gain β in a BJT operating in the active region.
- Q3. What is the function of the 'IOREF' and 'AREF' pins on an Arduino UNO board?
- Q4. Explain the photoconductive principle of a Cadmium Sulfide (CdS) LDR.
- Q5. Express decimal 29_{10} in (a) Pure Binary, and (b) BCD (8421).
- Q6. Why is Gray code called a "reflected" code?
- Q7. Write the dual of the Boolean algebraic identity: $A + \overline{A}B = A + B$.
- Q8. Draw the logic symbol and write the Boolean output equation for a 2-input XNOR gate.
- Q9. What is the primary operational difference between a latch and an edge-triggered flip-flop?
- Q10. Explain how an asynchronous counter can be truncated to count modulo-6.

SECTION B: Analytical & Numerical Problems ($6 \times 5 = 30$ Marks)

- Q11. In a common-emitter NPN transistor circuit, $V_{CC} = 15\text{ V}$, $R_C = 2.2\text{ k}\Omega$, $\beta = 100$. If $I_B = 50\text{ }\mu\text{A}$, calculate I_C , I_E and V_{CE} . Verify whether the transistor is in active or saturation mode.
- Q12. An Arduino UNO ADC pin receives a signal from an ultrasonic sensor. If the measured voltage is 2.45 V, calculate the 10-bit integer returned by 'analogRead()'.
- Q13. Convert Gray code 101101_{Gray} into its binary equivalent.
- Q14. Using Boolean algebra laws, simplify the expression: $F = \overline{A}\overline{B}C + \overline{A}B\overline{C} + ABC + A\overline{B}\overline{C}$.
- Q15. Design a 1-bit binary Half Adder using only 2-input NOR gates.
- Q16. Implement a Full Subtractor using a 3-to-8 line decoder and external logic gates.
- Q17. Convert a JK Flip-Flop into an SR Flip-Flop. Show the conversion table and K-Maps.

SECTION C: Comprehensive Design Questions ($5 \times 10 = 50$ Marks)

- Q18.** (a) Explain the working of a Full-Wave Center-Tapped Rectifier with a complete circuit diagram and waveforms.
 (b) Derive expressions for its DC output voltage, RMS voltage, ripple factor, and rectification efficiency.
 (c) Why is the PIV rating of diodes in a center-tapped rectifier twice that of a bridge rectifier?
- Q19.** (a) Detail the pin configuration of the Arduino UNO board, grouping pins by power, analog, digital I/O, PWM, and communications.
 (b) Explain how PWM approximates analog output voltages on pins 3, 5, 6, 9, 10, 11. Calculate the average output voltage for an 'analogWrite(6, 153)' command.
- Q20.** (a) Minimize the Boolean function using a 4-variable K-Map:

$$F(A, B, C, D) = \prod M(0, 1, 4, 5, 8, 9, 12, 13) \cdot d(3, 11)$$

 (b) Realize the minimal POS expression using only NOR gates.
- Q21.** (a) Explain the architecture and operation of a 4-bit Serial-In Parallel-Out (SIPO) shift register. Draw its schematic and timing diagram.
 (b) Explain the operational difference between SISO, SIPO, PISO, and PIPO shift registers.
- Q22.** Design a Synchronous 3-bit DOWN Counter ($7 \rightarrow 6 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0 \rightarrow 7$) using T Flip-Flops.
 (a) Formulate the State Table and determine T_2, T_1, T_0 excitation inputs.
 (b) Minimize input expressions using K-Maps.
 (c) Draw the complete logic circuit diagram.

9.5 Solutions for Mock Examination Paper 2**Selected Solutions to Paper 2****Q11 Solution (BJT Bias):**

$$I_C = \beta I_B = 100 \times 50 \mu\text{A} = 5.0 \text{ mA}$$

$$I_E = I_B + I_C = 0.05 + 5.0 = 5.05 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C R_C = 15 - (5.0 \text{ mA} \times 2.2 \text{ k}\Omega) = 15 - 11.0 = 4.0 \text{ V}$$

Since $V_{CE} = 4.0 \text{ V} > V_{CE(sat)} \approx 0.2 \text{ V}$, the transistor is in ****Active Mode****.

Q13 Solution (Gray to Binary): Gray = 101101₂:

- $B_5 = G_5 = 1$
- $B_4 = B_5 \oplus G_4 = 1 \oplus 0 = 1$
- $B_3 = B_4 \oplus G_3 = 1 \oplus 1 = 0$
- $B_2 = B_3 \oplus G_2 = 0 \oplus 1 = 1$
- $B_1 = B_2 \oplus G_1 = 1 \oplus 0 = 1$
- $B_0 = B_1 \oplus G_0 = 1 \oplus 1 = 0$

Binary = 110110_2 .

Q19 (b) Solution (PWM Average Voltage):

$$\text{Duty Cycle} = \frac{153}{255} = 0.60 \implies V_{avg} = 5.0 \text{ V} \times 0.60 = 3.0 \text{ V}$$

Q22 Solution (3-Bit Synchronous DOWN Counter): Toggle logic for down counter:
A flip-flop toggles when all lower-order bits are 0 ($\bar{Q} = 1$):

$$T_0 = 1, \quad T_1 = \bar{Q}_0, \quad T_2 = \bar{Q}_0 \cdot \bar{Q}_1$$